

9

SEMIGROUPS AND GROUPS

Prerequisites: Chapters 4, 5, and 6

The notion of a mathematical structure was introduced in Section 1.6. In the following chapters, other types of mathematical systems were developed; some, such as [propositions, $\wedge$, $\vee$, $\sim$], were not given specific names, but others, such as B_n , the Boolean algebra on n elements, were named. In this chapter, we identify two more types of mathematical structures, semigroups and groups. Semigroups will be used in our study of finite state machines in Chapter 10. We also develop the basic ideas of group theory, which we will apply to coding theory in Chapter 11.

9.1 BINARY OPERATIONS REVISITED

We defined binary operations earlier (see Section 1.6) and noted in Section 5.2 that a binary operation may be used to define a function. Here we turn the process around and define a binary operation as a function with certain properties.

A **binary operation on a set A** is an everywhere defined function $f: A \times A \rightarrow A$. Observe the following properties that a binary operation must satisfy:

1. Since $\text{Dom}(f) = A \times A$, f assigns an element $f(a, b)$ of A to each ordered pair (a, b) in $A \times A$. That is, the binary operation must be defined for each ordered pair of elements of A .
2. Since a binary operation is a function, only one element of A is assigned to each ordered pair.

Thus we can say that a binary operation is a rule that assigns to each ordered pair of elements of A a unique element of A . The reader should note that this definition is more restrictive than that given in Chapter 1, but we have made the change to simplify the discussion in this chapter. We shall now turn to a number of examples.

It is customary to denote binary operations by a symbol such as $*$, instead of f , and to denote the element assigned to (a, b) by $a * b$ [instead of $f(a, b)$]. It should be emphasized that if a and b are elements in A , then $a * b \in A$, and this property is often described by saying that A is **closed** under the operation $*$.

EXAMPLE 1 Let $A = \mathbb{Z}$. Define $a * b$ as $a + b$. Then $*$ is a binary operation on $\mathbb{Z}$.

EXAMPLE 2 Let $A = \mathbb{R}$. Define $a * b$ as a/b . Then $*$ is not a binary operation, since it is not defined for every ordered pair of elements of A . For example, $3 * 0$ is not defined, since we cannot divide by zero.

EXAMPLE 3 Let $A = \mathbb{Z}^+$. Define $a * b$ as $a - b$. Then $*$ is not a binary operation since it does not assign an element of A to every ordered pair of elements of A ; for example, $2 * 5 \notin A$.

EXAMPLE 4 Let $A = \mathbb{Z}$. Define $a * b$ as a number less than both a and b . Then $*$ is not a binary operation, since it does not assign a *unique* element of A to each ordered pair of elements of A ; for example, $8 * 6$ could be 5, 4, 3, 1, and so on. Thus, in this case, $*$ would be a relation from $A \times A$ to A , but not a function.

EXAMPLE 5 Let $A = \mathbb{Z}$. Define $a * b$ as $\max\{a, b\}$. Then $*$ is a binary operation; for example, $2 * 4 = 4$, $-3 * (-5) = -3$.

EXAMPLE 6 Let $A = P(S)$, for some set S . If V and W are subsets of S , define $V * W$ as $V \cup W$. Then $*$ is a binary operation on A . Moreover, if we define $V *' W$ as $V \cap W$, then $*'$ is another binary operation on A .

As Example 6 shows, it is possible to define many binary operations on the same set.

EXAMPLE 7 Let M be the set of all $n \times n$ Boolean matrices for a fixed n . Define $A * B$ as $A \vee B$ (see Section 1.5). Then $*$ is a binary operation. This is also true of $A \wedge B$.

EXAMPLE 8 Let L be a lattice. Define $a * b$ as $a \wedge b$ (the greatest lower bound of a and b). Then $*$ is a binary operation on L . This is also true of $a \vee b$ (the least upper bound of a and b).

Tables

If $A = \{a_1, a_2, \dots, a_n\}$ is a finite set, we can define a binary operation on A by means of a table as shown in Figure 9.1. The entry in position i, j denotes the element $a_i * a_j$.

EXAMPLE 9 Let $A = \{0, 1\}$. We define binary operations $\vee$ and $\wedge$ by the following tables:

$\vee$	0	1
0	0	1
1	1	1

$\wedge$	0	1
0	0	0
1	0	1

*	a_1	a_2	$\cdots$	a_j	$\cdots$	a_n
a_1						
a_2						
$\vdots$						
a_i				$a_i * a_j$		
$\vdots$						
a_n						

Figure 9.1

If $A = \{a, b\}$, we shall now determine the number of binary operations that can be defined on A . Every binary operation $*$ on A can be described by a table

*	a	b
a		
b		

Since every blank can be filled in with the element a or b , we conclude that there are $2 \cdot 2 \cdot 2 \cdot 2 = 2^4$ or 16 ways to complete the table. Thus, there are 16 binary operations on A .

Properties of Binary Operations

Several of the properties defined for binary operations in Section 1.6 are of particular importance in this chapter. We repeat them here.

A binary operation on a set A is said to be **commutative** if

$$a * b = b * a$$

for all elements a and b in A .

EXAMPLE 10

The binary operation of addition on Z (as discussed in Example 1) is commutative. ■

EXAMPLE 11

The binary operation of subtraction on Z is not commutative, since

$$2 - 3 \neq 3 - 2. \quad \blacksquare$$

A binary operation that is described by a table is commutative if and only if the entries in the table are symmetric with respect to the main diagonal.

EXAMPLE 12

Which of the following binary operations on $A = \{a, b, c, d\}$ are commutative?

*	a	b	c	d
a	a	c	b	d
b	b	c	b	a
c	c	d	b	c
d	a	a	b	b

(a)

*	a	b	c	d
a	a	c	b	d
b	c	d	b	a
c	b	b	a	c
d	d	a	c	d

(b)

Solution The operation in (a) is not commutative, since $a * b$ is c while $b * a$ is b . The operation in (b) is commutative, since the entries in the table are symmetric with respect to the main diagonal. ■

A binary operation $*$ on a set A is said to be **associative** if

$$a * (b * c) = (a * b) * c$$

for all elements a , b , and c in A .

EXAMPLE 13

The binary operation of addition on Z is associative. ■

EXAMPLE 14

The binary operation of subtraction on Z is not associative, since

$$2 - (3 - 5) \neq (2 - 3) - 5.$$

EXAMPLE 15

Let L be a lattice. The binary operation defined by $a * b = a \wedge b$ (see Example 8) is commutative and associative. It also satisfies the **idempotent** property $a \wedge a = a$. A partial converse of this example is also true, as shown in Example 16. ■

EXAMPLE 16

Let $*$ be a binary operation on a set A , and suppose that $*$ satisfies the following properties for any a , b , and c in A .

- $a = a * a$ Idempotent property
- $a * b = b * a$ Commutative property
- $a * (b * c) = (a * b) * c$ Associative property

Define a relation $\leq$ on A by

$$a \leq b \text{ if and only if } a = a * b.$$

Show that $(A, \leq)$ is a poset, and for all a, b in A , $\text{GLB}(a, b) = a * b$.

Solution We must show that $\leq$ is reflexive, antisymmetric, and transitive. Since $a = a * a$, $a \leq a$ for all a in A , and $\leq$ is reflexive.

Now suppose that $a \leq b$ and $b \leq a$. Then, by definition and property 2, $a = a * b = b * a = b$, so $a = b$. Thus $\leq$ is antisymmetric.

If $a \leq b$ and $b \leq c$, then $a = a * b = a * (b * c) = (a * b) * c = a * c$, so $a \leq c$ and $\leq$ is transitive.

Finally, we must show that, for all a and b in A , $a * b = a \wedge b$ (the greatest lower bound of a and b with respect to $\leq$). We have $a * b = a * (b * b) = (a * b) * b$, so $a * b \leq b$. In a similar way, we can show that $a * b \leq a$, so $a * b$ is a lower bound for a and b . Now, if $c \leq a$ and $c \leq b$, then $c = c * a$ and $c = c * b$ by definition. Thus $c = (c * a) * b = c * (a * b)$, so $c \leq a * b$. This shows that $a * b$ is the greatest lower bound of a and b . ■

9.1 Exercises

In Exercises 1 through 8, determine whether the description of $*$ is a valid definition of a binary operation on the set.

- On $\mathbb{R}$, where $a * b$ is ab (ordinary multiplication).
- On $\mathbb{Z}^+$, where $a * b$ is a/b .
- On $\mathbb{Z}$, where $a * b$ is a^b .
- On $\mathbb{Z}^+$, where $a * b$ is a^b .
- On $\mathbb{Z}^+$, where $a * b$ is $a - b$.
- On $\mathbb{R}$, where $a * b$ is $a\sqrt{b}$.
- On $\mathbb{R}$, where $a * b$ is the largest rational number that is less than ab .
- On $\mathbb{Z}$, where $a * b$ is $2a + b$.

In Exercises 9 through 17, determine whether the binary operation $*$ is commutative and whether it is associative on the set.

- On $\mathbb{Z}^+$, where $a * b$ is $a + b + 2$.
- On $\mathbb{Z}$, where $a * b$ is ab .
- On $\mathbb{R}$, where $a * b$ is $a \times |b|$.
- On the set of nonzero real numbers, where $a * b$ is a/b .
- On $\mathbb{R}$, where $a * b$ is the minimum of a and b .
- On the set of $n \times n$ Boolean matrices, where $A * B$ is $A \odot B$ (see Section 1.5).
- On $\mathbb{R}$, where $a * b$ is $ab/3$.
- On $\mathbb{R}$, where $a * b$ is $ab + 2b$.
- On a lattice A , where $a * b$ is $a \vee b$.
- Fill in the following table so that the binary operation $*$ is commutative.

$*$	a	b	c
a			
b			
c			

- Fill in the following table so that the binary operation $*$ is commutative and has the idempotent property.

$*$	a	b	c
a			
b			
c			

- Consider the binary operation $*$ defined on the set $A = \{a, b, c\}$ by the following table.

$*$	a	b	c
a			
b			
c			

- Is $*$ a commutative operation?
- Compute $a * (b * c)$ and $(a * b) * c$.
- Is $*$ an associative operation?

- Consider the binary operation $*$ defined on the set $A = \{a, b, c, d\}$ by the following table.

$*$	a	b	c	d
a				
b				
c				
d				

Compute

- $c * d$ and $d * c$.
- $b * d$ and $d * b$.
- $a * (b * c)$ and $(a * b) * c$.
- Is $*$ commutative? associative?

In Exercises 22 and 23, complete the given table so that the binary operation $*$ is associative.

- | $*$ | a | b | c | d |
|-----|-----|-----|-----|-----|
| a | | | | |
| b | | | | |
| c | | | | |
| d | | | | |

23.

*	a	b	c	d
a	b	a	c	d
b	b	a	c	d
c				
d	d	c	c	d

24. Let A be a set with n elements. How many binary operations can be defined on A ?

25. Let A be a set with n elements. How many commutative binary operations can be defined on A ?

26. Let $A = \{a, b\}$.

- Make a table for each of the 16 binary operations that can be defined on A .
- Using part (a), identify the binary operations on A that are commutative.

27. Let $A = \{a, b\}$.

- Using Exercise 26, identify the binary operations on A that are associative.
- Using Exercise 26, identify the binary operations on A that satisfy the idempotent property.

28. Let $*$ be a binary operation on a set A , and suppose that $*$ satisfies the idempotent, commutative, and associative properties, as discussed in Example 16. Define a relation $\leq$ on A by $a \leq b$ if and only if $b = a * b$. Show that $(A, \leq)$ is a poset and, for all a and b , $\text{LUB}(a, b) = a * b$.

29. Describe how the definition of a binary operation on a set A is different from the definition of a binary operation given in Section 1.6. Explain also whether a binary operation on a set is or is not a binary operation according to the earlier definition.

9.2 SEMIGROUPS

In this section we define a simple mathematical structure, consisting of a set together with a binary operation, that has many important applications.

A **semigroup** is a nonempty set S together with an **associative** binary operation $*$ defined on S . We shall denote the semigroup by $(S, *)$ or, when it is clear what the operation $*$ is, simply by S . We also refer to $a * b$ as the **product** of a and b . The semigroup $(S, *)$ is said to be commutative if $*$ is a commutative operation.

EXAMPLE 1

It follows from Section 9.1 that $(Z, +)$ is a commutative semigroup.

EXAMPLE 2

The set $P(S)$, where S is a set, together with the operation of union is a commutative semigroup.

EXAMPLE 3

The set Z with the binary operation of subtraction is not a semigroup, since subtraction is not associative.

EXAMPLE 4

Let S be a fixed nonempty set, and let S^S be the set of all functions $f: S \rightarrow S$. If f and g are elements of S^S , we define $f * g$ as $f \circ g$, the composite function. Then $*$ is a binary operation on S^S , and it follows from Section 4.7 that $*$ is associative. Hence $(S^S, *)$ is a semigroup. The semigroup S^S is not commutative.

EXAMPLE 5

Let $(L, \leq)$ be a lattice. Define a binary operation on L by $a * b = a \vee b$. Then L is a semigroup.

EXAMPLE 6

Let $A = \{a_1, a_2, \dots, a_n\}$ be a nonempty set. Recall from Section 1.3 that A^* is the set of all finite sequences of elements of A . That is, A^* consists of all words that can be formed from the alphabet A . Let α and β be elements of A^* . Observe that catenation is a binary operation $\cdot$ on A^* . Recall that if $\alpha = a_1 a_2 \dots a_n$ and

$\beta = b_1 b_2 \cdots b_k$, then $\alpha \cdot \beta = a_1 a_2 \cdots a_n b_1 b_2 \cdots b_k$. It is easy to see that if α , β , and γ are any elements of A^* , then

$$\alpha \cdot (\beta \cdot \gamma) = (\alpha \cdot \beta) \cdot \gamma$$

so that $\cdot$ is an associative binary operation, and $(A^*, \cdot)$ is a semigroup. The semigroup $(A^*, \cdot)$ is called the **free semigroup generated by A** .

In a semigroup $(S, *)$ we can establish the following generalization of the associative property; we omit the proof.

Theorem 1 If $a_1, a_2, \dots, a_n$, $n \geq 3$, are arbitrary elements of a semigroup, then all products of the elements $a_1, a_2, \dots, a_n$ that can be formed by inserting meaningful parentheses arbitrarily are equal.

EXAMPLE 7

Theorem 1 shows that the products

$$((a_1 * a_2) * a_3) * a_4, \quad a_1 * (a_2 * (a_3 * a_4)), \quad (a_1 * (a_2 * a_3)) * a_4$$

are all equal.

If $a_1, a_2, \dots, a_n$ are elements in a semigroup $(S, *)$, we shall write their product as

$$a_1 * a_2 * \cdots * a_n,$$

omitting the parentheses.

An element e in a semigroup $(S, *)$ is called an **identity element** if

$$e * a = a * e = a$$

for all $a \in S$. As shown by Theorem 1, Section 1.6, an identity element must be unique.

EXAMPLE 8

The number 0 is an identity in the semigroup $(\mathbb{Z}, +)$.

EXAMPLE 9

The semigroup $(\mathbb{Z}^+, +)$ has no identity element.

A **monoid** is a semigroup $(S, *)$ that has an identity.

EXAMPLE 10

The semigroup $P(S)$ defined in Example 2 has the identity $\emptyset$, since

$$\emptyset * A = \emptyset \cup A = A = A \cup \emptyset = A * \emptyset$$

for any element $A \in P(S)$. Hence $P(S)$ is a monoid.

EXAMPLE 11

The semigroup S^S defined in Example 4 has the identity 1_S , since

$$1_S * f = 1_S \circ f = f \circ 1_S = f * 1_S$$

for any element $f \in S^S$, we see that S^S is a monoid.

EXAMPLE 12

The semigroup A^* defined in Example 6 is actually a monoid with identity Λ , the empty sequence, since $\alpha \cdot \Lambda = \Lambda \cdot \alpha = \alpha$ for all $\alpha \in A^*$.

EXAMPLE 13

The set of all relations on a set A is a monoid under the operation of composition. The identity element is the equality relation Δ (see Section 4.7).

Let $(S, *)$ be a semigroup and let T be a subset of S . If T is closed under the operation $*$ (that is, $a * b \in T$ whenever a and b are elements of T), then $(T, *)$ is called a **subsemigroup** of $(S, *)$. Similarly, let $(S, *)$ be a monoid with identity e , and let T be a nonempty subset of S . If T is closed under the operation $*$ and $e \in T$, then $(T, *)$ is called a **submonoid** of $(S, *)$.

Observe that the associative property holds in any subset of a semigroup so that a subsemigroup $(T, *)$ of a semigroup $(S, *)$ is itself a semigroup. Similarly, a submonoid of a monoid is itself a monoid.

EXAMPLE 14

If T is the set of all even integers, then $(T, \times)$ is a subsemigroup of the monoid $(\mathbb{Z}, \times)$, where $\times$ is ordinary multiplication, but it is not a submonoid since the identity of $\mathbb{Z}$, the number 1, does not belong to T .

EXAMPLE 15

If $(S, *)$ is a semigroup, then $(S, *)$ is a subsemigroup of $(S, *)$. Similarly, let $(S, *)$ be a monoid. Then $(S, *)$ is a submonoid of $(S, *)$, and if $T = \{e\}$, then $(T, *)$ is also a submonoid of $(S, *)$.

Suppose that $(S, *)$ is a semigroup, and let $a \in S$. For $n \in \mathbb{Z}^+$, we define the powers of a^n recursively as follows:

$$a^1 = a, \quad a^n = a^{n-1} * a, \quad n \geq 2.$$

Moreover, if $(S, *)$ is a monoid, we also define

$$a^0 = e.$$

It can be shown that if m and n are nonnegative integers, then

$$a^m * a^n = a^{m+n}.$$

EXAMPLE 16

(a) If $(S, *)$ is a semigroup, $a \in S$, and

$$T = \{a^i \mid i \in \mathbb{Z}^+\},$$

then $(T, *)$ is a subsemigroup of $(S, *)$.

(b) If $(S, *)$ is a monoid, $a \in S$, and

$$T = \{a^i \mid i \in \mathbb{Z}^+ \text{ or } i = 0\},$$

then $(T, *)$ is a submonoid of $(S, *)$.

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Isomorphism and Homomorphism

An isomorphism between two posets was defined in Section 6.1 as a one-to-one correspondence that preserved order relations, the distinguishing feature of posets. We now define an isomorphism between two semigroups as a one-to-one correspondence that preserves the binary operations. In general, an isomorphism between two mathematical structures of the same type should preserve the distinguishing features of the structures.

Let $(S, *)$ and $(T, \circ)$ be two semigroups. A function $f: S \rightarrow T$ is called an **isomorphism** from $(S, *)$ to $(T, \circ)$ if it is a one-to-one correspondence from S to T , and if

$$f(a * b) = f(a) \circ f(b)$$

for all a and b in S .

If f is an isomorphism from $(S, *)$ to $(T, \circ)$, then, since f is a one-to-one correspondence, it follows from Theorem 1 of Section 5.1 that f^{-1} exists and is a one-to-one correspondence from T to S . We now show that f^{-1} is an isomorphism from $(T, \circ)$ to $(S, *)$. Let a' and b' be any elements of T . Since f is onto, we can find elements a and b in S such that $f(a) = a'$ and $f(b) = b'$. Then $a = f^{-1}(a')$ and $b = f^{-1}(b')$. Now

$$\begin{aligned} f^{-1}(a' \circ b') &= f^{-1}(f(a) \circ f(b)) \\ &= f^{-1}(f(a * b)) \\ &= (f^{-1} \circ f)(a * b) \\ &= a * b = f^{-1}(a') * f^{-1}(b'). \end{aligned}$$

Hence f^{-1} is an isomorphism.

We now say that the semigroups $(S, *)$ and $(T, \circ)$ are **isomorphic** and we write $S \simeq T$.

To show that two semigroups $(S, *)$ and $(T, \circ)$ are isomorphic, we use the following procedure:

STEP 1: Define a function $f: S \rightarrow T$ with $\text{Dom}(f) = S$.

STEP 2: Show that f is one-to-one.

STEP 3: Show that f is onto.

STEP 4: Show that $f(a * b) = f(a) \circ f(b)$.

EXAMPLE 17

Let T be the set of all even integers. Show that the semigroups $(Z, +)$ and $(T, +)$ are isomorphic.

Solution

STEP 1: We define the function $f: Z \rightarrow T$ by $f(a) = 2a$.

STEP 2: We now show that f is one-to-one as follows. Suppose that $f(a_1) = f(a_2)$. Then $2a_1 = 2a_2$, so $a_1 = a_2$. Hence f is one-to-one.

STEP 3: We next show that f is onto. Suppose that b is any even integer. Then $a = b/2 \in Z$ and

$$f(a) = f(b/2) = 2(b/2) = b,$$

so f is onto.

STEP 4: We have

$$f(a+b) = 2(a+b) = 2a+2b = f(a) + f(b).$$

Hence $(Z, +)$ and $(T, +)$ are isomorphic semigroups. ■

In general, it is rather straightforward to verify that a given function $f: S \rightarrow T$ is or is not an isomorphism. However, it is generally more difficult to show that two semigroups are isomorphic, because one has to create the isomorphism f .

As in the case of poset or lattice isomorphisms, when two semigroups $(S, *)$ and $(T, *)'$ are isomorphic, they can differ only in the nature of their elements; their semigroup structures are identical. If S and T are finite semigroups, their respective binary operations can be given by tables. Then S and T are isomorphic if we can rearrange and relabel the elements of S so that its table is identical with that of T .

EXAMPLE 18

Let $S = \{a, b, c\}$ and $T = \{x, y, z\}$. It is easy to verify that the following operation tables give semigroup structures for S and T , respectively.

$*$	a	b	c
a	a	b	c
b	b	c	a
c	c	a	b

$*$	x	y	z
x	z	x	y
y	x	y	z
z	y	z	x

Let

$$f(a) = y$$

$$f(b) = x$$

$$f(c) = z.$$

Replacing the elements in S by their images and rearranging the table, we obtain exactly the table for T . Thus S and T are isomorphic. ■

Theorem 2 Let $(S, *)$ and $(T, *')$ be monoids with identities e and e' , respectively. Let $f: S \rightarrow T$ be an isomorphism. Then $f(e) = e'$.

Proof Let b be any element of T . Since f is onto, there is an element a in S such that $f(a) = b$. Then

$$\begin{aligned} a &= a * e \\ b = f(a) &= f(a * e) = f(a) *' f(e) \\ &= b *' f(e). \end{aligned}$$

Similarly, since $a = e * a$, $b = f(e) *' b$. Thus for any $b \in T$,

$$b = b *' f(e) = f(e) *' b,$$

which means that $f(e)$ is an identity for T . Thus since the identity is unique, it follows that $f(e) = e'$.

If $(S, *)$ and $(T, *')$ are semigroups such that S has an identity and T does not, it then follows from Theorem 2 that $(S, *)$ and $(T, *')$ cannot be isomorphic.

EXAMPLE 19

Let T be the set of all even integers and let $\times$ be ordinary multiplication. Then the semigroups $(Z, \times)$ and $(T, \times)$ are not isomorphic, since Z has an identity and T does not. ■

By dropping the conditions of one to one and onto in the definition of an isomorphism of two semigroups, we get another important method for comparing the algebraic structures of the two semigroups.

Let $(S, *)$ and $(T, \cdot)$ be two semigroups. An everywhere-defined function $f: S \rightarrow T$ is called a **homomorphism** from $(S, *)$ to $(T, \cdot)$ if

$$f(a * b) = f(a) \cdot f(b)$$

for all a and b in S . If f is also onto, we say that T is a **homomorphic image** of S .

EXAMPLE 20

Let $A = \{0, 1\}$ and consider the semigroups $(A^*, \cdot)$ and $(A, +)$, where $\cdot$ is the concatenation operation and $+$ is defined by the table

+	0	1
0	0	1
1	1	0

Define the function $f: A^* \rightarrow A$ by

$$f(\alpha) = \begin{cases} 1 & \text{if } \alpha \text{ has an odd number of 1's} \\ 0 & \text{if } \alpha \text{ has an even number of 1's} \end{cases}$$

It is easy to verify that if α and β are any elements of A^* , then

$$f(\alpha \cdot \beta) = f(\alpha) + f(\beta).$$

Thus f is a homomorphism. The function f is onto since

$$f(0) = 0$$

$$f(1) = 1$$

but f is not an isomorphism, since it is not one to one. ■

The difference between an isomorphism and a homomorphism is that an isomorphism must be one to one and onto. For both an isomorphism and a homomorphism, the image of a product is the product of the images.

The proof of the following theorem, which is left as an exercise for the reader, is completely analogous to the proof of Theorem 2.

Theorem 3

Let $(S, *)$ and $(T, \cdot)$ be monoids with identities e and e' , respectively. Let $f: S \rightarrow T$ be a homomorphism from $(S, *)$ onto $(T, \cdot)$. Then $f(e) = e'$. ■

Theorem 3 is a stronger, or more general, statement than Theorem 2, because it requires fewer (weaker) conditions for the conclusion.

Theorem 3, together with the following two theorems, shows that, if a semigroup $(T, \cdot)$ is a homomorphic image of the semigroup $(S, *)$, then $(T, \cdot)$ has a strong algebraic resemblance to $(S, *)$.

Theorem 4 Let f be a homomorphism from a semigroup $(S, *)$ to a semigroup $(T, \cdot)$. If S' is a subsemigroup of $(S, *)$, then

$$f(S') = \{t \in T \mid t = f(s) \text{ for some } s \in S'\},$$

the image of S' under f , is a subsemigroup of $(T, \cdot)$.

Proof If t_1 and t_2 are any elements of $f(S')$, then there exist s_1 and s_2 in S' with

$$t_1 = f(s_1) \quad \text{and} \quad t_2 = f(s_2).$$

Then

$$t_1 \cdot t_2 = f(s_1) \cdot f(s_2) = f(s_1 * s_2) = f(s_3),$$

where $s_3 = s_1 * s_2 \in S'$. Hence $t_1 \cdot t_2 \in f(S')$.

Thus $f(S')$ is closed under the operation $\cdot$. Since the associative property holds in T , it holds in $f(S')$, so $f(S')$ is a subsemigroup of $(T, \cdot)$.

Theorem 5 If f is a homomorphism from a commutative semigroup $(S, *)$ onto a semigroup $(T, \cdot)$, then $(T, \cdot)$ is also commutative.

Proof Let t_1 and t_2 be any elements of T . Then there exist s_1 and s_2 in S with

$$t_1 = f(s_1) \quad \text{and} \quad t_2 = f(s_2).$$

Therefore,

$$t_1 \cdot t_2 = f(s_1) \cdot f(s_2) = f(s_1 * s_2) = f(s_2 * s_1) = f(s_2) \cdot f(s_1) = t_2 \cdot t_1.$$

Hence $(T, \cdot)$ is also commutative.

Exercises

1. Let $A = \{a, b\}$. Which of the following tables define a semigroup on A ? Which define a monoid on A ?

(a)

*	a	b
a	a	b
b	a	a

(b)

*	a	b
a	a	b
b	b	b

2. Let $A = \{a, b\}$. Which of the following tables define a semigroup on A ? Which define a monoid on A ?

(a)

*	a	b
a	b	a
b	a	b

(b)

*	a	b
a	a	b
b	b	a

3. Let $A = \{a, b\}$. Which of the following tables define a semigroup on A ? Which define a monoid on A ?

(a)

*	a	b
a	a	a
b	b	b

(b)

*	a	b
a	b	b
b	a	a

In Exercises 4 through 14, determine whether the set together with the binary operation is a semigroup, a monoid, or neither. If it is a monoid, specify the identity. If it is a semigroup or monoid, determine if it is commutative.

- $\mathbb{Z}^+$, where $*$ is defined as ordinary multiplication.
- $\mathbb{Z}^+$, where $a * b$ is defined as $\max\{a, b\}$.
- $\mathbb{Z}^+$, where $a * b$ is defined as $\text{GCD}\{a, b\}$.
- $\mathbb{Z}^+$, where $a * b$ is defined as a .
- The nonzero real numbers, where $*$ is ordinary multiplication.
- $P(S)$, with S a set, where $*$ is defined as intersection.
- A Boolean algebra B , where $a * b$ is defined as $a \wedge b$.

11. $S = \{1, 2, 3, 6, 12\}$, where $a * b$ is defined as $\text{GCD}(a, b)$.

12. $S = \{1, 2, 3, 6, 9, 18\}$, where $a * b$ is defined as $\text{LCM}(a, b)$.

13. Z , where $a * b = a + b - ab$.

14. The even integers, where $a * b$ is defined as $\frac{ab}{2}$.

15. Does the following table define a semigroup?

$*$	a	b	c
a	c	b	a
b	b	c	b
c	a	b	c

16. Does the following table define a semigroup?

$*$	a	b	c
a	a	c	b
b	c	b	a
c	b	a	c

17. Complete the following table to obtain a semigroup.

$*$	a	b	c
a	c	a	b
b	a	b	c
c			a

18. Complete the following table so that it defines a monoid.

$*$	a	b	c	d
a	c	d	a	b
b		a	b	
c			c	
d	b		d	a

19. Let $S = \{a, b\}$. Write the operation table for the semigroup S^* . Is the semigroup commutative?

20. Let $S = \{a, b\}$. Write the operation table for the semigroup $(P(S), \cup)$.

21. Let $A = \{a, b, c\}$ and consider the semigroup $(A^*, \cdot)$, where $\cdot$ is the operation of catenation. If $\alpha = abac$, $\beta = cba$, and $\gamma = babc$, compute

(a) $(\alpha\beta)\cdot\gamma$ (b) $\gamma\cdot(\alpha\alpha)$ (c) $(\gamma\beta)\cdot\alpha$

22. Prove or disprove that the intersection of two subsemigroups of a semigroup $(S, *)$ is a subsemigroup of $(S, *)$.

23. Prove or disprove that the intersection of two submonoids of a monoid $(S, *)$ is a submonoid of $(S, *)$.

24. Let $A = \{0, 1\}$, and consider the semigroup $(A^*, \cdot)$, where $\cdot$ is the operation of catenation. Let T be the subset of A^* consisting of all sequences having an odd number of 1's. Is $(T, \cdot)$ a subsemigroup of $(A, \cdot)$?

25. Let $A = \{a, b\}$. Are there two semigroups $(A, *)$ and $(A, \cdot)$ that are not isomorphic?

26. An element x in a monoid is called an **idempotent** if $x^2 = x * x = x$. Show that the set of all idempotents in a commutative monoid S is a submonoid of S .

27. Let $(S_1, *)$, $(S_2, *)$, and $(S_3, *)$ be semigroups and $f: S_1 \rightarrow S_2$ and $g: S_2 \rightarrow S_3$ be homomorphisms. Prove that $g \circ f$ is a homomorphism from S_1 to S_3 .

28. Let $(S_1, *)$, $(S_2, *)$, and $(S_3, *)$ be semigroups, and let $f: S_1 \rightarrow S_2$ and $g: S_2 \rightarrow S_3$ be isomorphisms. Show that $g \circ f: S_1 \rightarrow S_3$ is an isomorphism.

29. Which properties of f are used in the proof of Theorem 2?

30. Explain why the proof of Theorem 1 can be used as a proof of Theorem 3.

31. Let R^+ be the set of all positive real numbers. Show that the function $f: R^+ \rightarrow R$ defined by $f(x) = \ln x$ is an isomorphism of the semigroup $(R^+, \times)$ to the semigroup $(R, +)$, where $\times$ and $+$ are ordinary multiplication and addition, respectively.

9.3 PRODUCTS AND QUOTIENTS OF SEMIGROUPS

In this section we shall obtain new semigroups from existing semigroups.

Theorem 1 If $(S, *)$ and $(T, \cdot)$ are semigroups, then $(S \times T, *)$ is a semigroup, where $*$ is defined by $(s_1, t_1) * (s_2, t_2) = (s_1 * s_2, t_1 \cdot t_2)$.

Proof The proof is left as an exercise.

It follows at once from Theorem 1 that if S and T are monoids with identities e_S and e_T , respectively, then $S \times T$ is a monoid with identity (e_S, e_T) .

We now turn to a discussion of equivalence relations on a semigroup $(S, *)$. Since a semigroup is not merely a set, we shall find that certain equivalence relations on a semigroup give additional information about the structure of the semigroup.

An equivalence relation R on the semigroup $(S, *)$ is called a **congruence relation** if

$$a R a' \text{ and } b R b' \text{ imply } (a * b) R (a' * b').$$

EXAMPLE 1

Consider the semigroup $(Z, +)$ and the equivalence relation R on Z defined by

$$a R b \text{ if and only if } a \equiv b \pmod{2}.$$

Recall that we discussed this equivalence relation in Section 4.5. Remember that if $a \equiv b \pmod{2}$, then $2 \mid (a - b)$. We now show that this relation is a congruence relation as follows.

If

$$a \equiv b \pmod{2} \text{ and } c \equiv d \pmod{2},$$

then 2 divides $a - b$ and 2 divides $c - d$, so

$$a - b = 2m \text{ and } c - d = 2n,$$

where m and n are in Z . Adding, we have

$$(a - b) + (c - d) = 2m + 2n$$

or

$$(a + c) - (b + d) = 2(m + n),$$

so

$$a + c \equiv b + d \pmod{2}.$$

Hence the relation is a congruence relation. ■

EXAMPLE 2

Let $A = \{0, 1\}$ and consider the free semigroup $(A^*, \cdot)$ generated by A . Define the following relation on A :

$$\alpha R \beta \text{ if and only if } \alpha \text{ and } \beta \text{ have the same number of 1's.}$$

Show that R is a congruence relation on $(A^*, \cdot)$.

Solution We first show that R is an equivalence relation. We have

1. $\alpha R \alpha$ for any $\alpha \in A^*$.
2. If $\alpha R \beta$, then α and β have the same number of 1's, so $\beta R \alpha$.
3. If $\alpha R \beta$ and $\beta R \gamma$, then α and β have the same number of 1's and β and γ have the same number of 1's, so α and γ have the same number of 1's. Hence $\alpha R \gamma$.

We next show that R is a congruence relation. Suppose that $\alpha R \alpha'$ and $\beta R \beta'$. Then α and α' have the same number of 1's and β and β' have the same number of 1's. Since the number of 1's in $\alpha \cdot \beta$ is the sum of the number

of 1's in α and the number of 1's in β , we conclude that the number of 1's in $\alpha \cdot \beta$ is the same as the number of 1's in $\alpha' \cdot \beta'$. Hence

$$(\alpha \cdot \beta) R (\alpha' \cdot \beta')$$

and thus R is a congruence relation. ■

EXAMPLE 3

Consider the semigroup $(Z, +)$, where $+$ is ordinary addition. Let $f(x) = x^2 - x - 2$. We now define the following relation on Z :

$$a R b \text{ if and only if } f(a) = f(b).$$

It is straightforward to verify that R is an equivalence relation on Z . However, R is not a congruence relation since we have

$$-1 R 2 \quad \text{since } f(-1) = f(2) = 0$$

and

$$-2 R 3 \quad \text{since } f(-2) = f(3) = 4$$

but

$$(-1 + -2) \not R (2 + 3)$$

$$\text{since } f(-3) = 10 \text{ and } f(5) = 18. \quad \blacksquare$$

Recall from Section 4.5 that an equivalence relation R on the semigroup $(S, *)$ determines a partition of S . We let $[a] = R(a)$ be the equivalence class containing a and S/R denote the set of all equivalence classes. The notation $[a]$ is more traditional in this setting and produces less confusing computations.

Theorem 2

Let R be a congruence relation on the semigroup $(S, *)$. Consider the relation $\otimes$ from $S/R \times S/R$ to S/R in which the ordered pair $([a], [b])$ is, for a and b in S , related to $[a * b]$.

(a) $\otimes$ is a function from $S/R \times S/R$ to S/R , and as usual we denote $\otimes([a], [b])$ by $[a] \otimes [b]$. Thus $[a] \otimes [b] = [a * b]$.

(b) $(S/R, \otimes)$ is a semigroup. ■

Proof Suppose that $([a], [b]) = ([a'], [b'])$. Then $a R a'$ and $b R b'$, so we must have $a * b R a' * b'$, since R is a congruence relation. Thus $[a * b] = [a' * b']$; that is, $\otimes$ is a function. This means that $\otimes$ is a binary operation on S/R .

Next, we must verify that $\otimes$ is an associative operation. We have

$$\begin{aligned} [a] \otimes ([b] \otimes [c]) &= [a] \otimes [b * c] \\ &= [a * (b * c)] \\ &= [(a * b) * c] \quad \text{by the associative property of } * \text{ in } S \\ &= [a * b] \otimes [c] \\ &= ([a] \otimes [b]) \otimes [c]. \end{aligned}$$

Hence S/R is a semigroup. We call S/R the **quotient semigroup** or **factor semigroup**. Observe that $\otimes$ is a type of "quotient binary relation" on S/R that is constructed from the original binary relation $*$ on S by the congruence relation R . ▼

Corollary 1

Let R be a congruence relation on the monoid $(S, *)$. If we define the operation $\oplus$ in S/R by $[a] \oplus [b] = [a * b]$, then $(S/R, \oplus)$ is a monoid.

Proof If e is the identity in $(S, *)$, then it is easy to verify that $[e]$ is the identity in $(S/R, \oplus)$.

EXAMPLE 4

Consider the situation in Example 2. Since R is a congruence relation on the monoid $S = (A^*, \cdot)$, we conclude that $(S/R, \odot)$ is a monoid, where

$$[\alpha] \odot [\beta] = [\alpha \cdot \beta].$$

EXAMPLE 5

As has already been pointed out in Section 4.5, we can repeat Example 4 of that section with the positive integer n instead of 2. That is, we define the following relation on the semigroup $(Z, +)$:

$$a R b \quad \text{if and only if} \quad a \equiv b \pmod{n}.$$

Using exactly the same method as in Example 4 in Section 4.5, we show that R is an equivalence relation and, as in the case of $n = 2$, $a \equiv b \pmod{n}$ implies $n \mid (a - b)$. Thus, if n is 4, then

$$2 \equiv 6 \pmod{4}$$

and 4 divides $(2 - 6)$. We also leave it for the reader to show that $\equiv \pmod{n}$ is a congruence relation on Z .

We now let $n = 4$ and we compute the equivalence classes determined by the congruence relation $\equiv \pmod{4}$ on Z . We obtain

$$[0] = \{\dots, -8, -4, 0, 4, 8, 12, \dots\} = [4] = [8] = \dots$$

$$[1] = \{\dots, -7, -3, 1, 5, 9, 13, \dots\} = [5] = [9] = \dots$$

$$[2] = \{\dots, -6, -2, 2, 6, 10, 14, \dots\} = [6] = [10] = \dots$$

$$[3] = \{\dots, -5, -1, 3, 7, 11, 15, \dots\} = [7] = [11] = \dots$$

These are all the distinct equivalence classes that form the quotient set $Z/\equiv \pmod{4}$. It is customary to denote the quotient set $Z/\equiv \pmod{n}$ by Z_n ; Z_n is a monoid with operation $\oplus$ and identity $[0]$. We now determine the addition table for the semigroup Z_4 with operation $\oplus$.

$\oplus$	[0]	[1]	[2]	[3]
[0]	[0]	[1]	[2]	[3]
[1]	[1]	[2]	[3]	[0]
[2]	[2]	[3]	[0]	[1]
[3]	[3]	[0]	[1]	[2]

The entries in this table are obtained from

$$[a] \oplus [b] = [a + b].$$

Thus

$$\begin{aligned}[1] \oplus [2] &= [1 + 2] = [3] \\ [1] \oplus [3] &= [1 + 3] = [4] = [0] \\ [2] \oplus [3] &= [2 + 3] = [5] = [1] \\ [3] \oplus [3] &= [3 + 3] = [6] = [2].\end{aligned}$$

It can be shown that, in general, Z_n has the n equivalence classes

$$[0], [1], [2], \dots, [n-1]$$

and that

$$[a] \oplus [b] = [r],$$

where r is the remainder when $a + b$ is divided by n . Thus, if n is 6,

$$\begin{aligned}[2] \oplus [3] &= [5] \\ [3] \oplus [5] &= [2] \\ [3] \oplus [3] &= [0].\end{aligned}$$

We shall now examine the connection between the structure of a semigroup $(S, *)$ and the quotient semigroup $(S/R, \oplus)$, where R is a congruence relation on $(S, *)$.

Theorem 3

Let R be a congruence relation on a semigroup $(S, *)$, and let $(S/R, \oplus)$ be the corresponding quotient semigroup. Then the function $f_R: S \rightarrow S/R$ defined by

$$f_R(a) = [a]$$

is an onto homomorphism, called the **natural homomorphism**.

Proof If $[a] \in S/R$, then $f_R(a) = [a]$, so f_R is an onto function. Moreover, if a and b are elements of S , then

$$f_R(a * b) = [a * b] = [a] \oplus [b] = f_R(a) \oplus f_R(b),$$

so f_R is a homomorphism.

Theorem 4
Fundamental
Homomorphism Theorem

Let $f: S \rightarrow T$ be a homomorphism of the semigroup $(S, *)$ onto the semigroup $(T, *)$. Let R be the relation on S defined by $a R b$ if and only if $f(a) = f(b)$, for a and b in S . Then

- R is a congruence relation.
- $(T, *)$ and the quotient semigroup $(S/R, \oplus)$ are isomorphic.

Proof

- (a) We show that R is an equivalence relation. First, $a R a$ for every $a \in S$, since $f(a) = f(a)$. Next, if $a R b$, then $f(a) = f(b)$, so $b R a$. Finally, if $a R b$ and $b R c$, then $f(a) = f(b)$ and $f(b) = f(c)$, so $f(a) = f(c)$ and $a R c$. Hence R is an equivalence relation. Now suppose that $a R a_1$ and $b R b_1$. Then

$$f(a) = f(a_1) \quad \text{and} \quad f(b) = f(b_1).$$

Multiplying in T , we obtain

$$f(a) *' f(b) = f(a_1) *' f(b_1).$$

Since f is a homomorphism, this last equation can be rewritten as

$$f(a * b) = f(a_1 * b_1).$$

Hence

$$(a * b) R (a_1 * b_1)$$

and R is a congruence relation.

- (b) We now consider the relation $\bar{f}$ from S/R to T defined as follows:

$$\bar{f} = \{([a], f(a)) \mid [a] \in S/R\}.$$

We first show that $\bar{f}$ is a function. Suppose that $[a] = [a']$. Then $a R a'$, so $f(a) = f(a')$, which implies that $\bar{f}$ is a function. We may now write $\bar{f}: S/R \rightarrow T$, where $\bar{f}([a]) = f(a)$ for $[a] \in S/R$.

We next show that $\bar{f}$ is one to one. Suppose that $\bar{f}([a]) = \bar{f}([a'])$. Then

$$f(a) = f(a').$$

So $a R a'$, which implies that $[a] = [a']$. Hence $\bar{f}$ is one to one.

Now we show that $\bar{f}$ is onto. Suppose that $b \in T$. Since f is onto, $f(a) = b$ for some element a in S . Then

$$\bar{f}([a]) = f(a) = b.$$

So $\bar{f}$ is onto.

Finally,

$$\begin{aligned} \bar{f}([a] \otimes [b]) &= \bar{f}([a * b]) \\ &= f(a * b) = f(a) *' f(b) \\ &= \bar{f}([a]) *' \bar{f}([b]). \end{aligned}$$

Hence $\bar{f}$ is an isomorphism.

EXAMPLE 6

Let $A = \{0, 1\}$, and consider the free semigroup A^* generated by A under the operation of concatenation. Note that A^* is a monoid with the empty string Λ as identity. Let N be the set of all nonnegative integers. Then N is a semigroup under the operation of ordinary addition, denoted by $(N, +)$. The function $f: A^* \rightarrow$

$$f(\alpha) = \text{the number of 1's in } \alpha$$

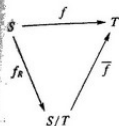


Figure 9.2

is readily checked to be a homomorphism. Let R be the following relation on A^* :

$$\alpha R \beta \quad \text{if and only if} \quad f(\alpha) = f(\beta).$$

That is, $\alpha R \beta$ if and only if α and β have the same number of 1's. Theorem 4 implies that $A^*/R \cong N$ under the isomorphism $\bar{f}: A^*/R \rightarrow N$ defined by

$$\bar{f}([\alpha]) = f(\alpha) = \text{the number of 1's in } \alpha.$$

Theorem 4(b) can be described by the diagram shown in Figure 9.2. Here f_R is the natural homomorphism. It follows from the definitions of f_R and $\bar{f}$ that

$$\bar{f} \circ f_R = f$$

since

$$(\bar{f} \circ f_R)(a) = \bar{f}(f_R(a)) = \bar{f}([a]) = f(a).$$

9.3 Exercises

- Let $(S, *)$ and $(T, \cdot)$ be commutative semigroups. Show that $S \times T$ (see Theorem 1) is also a commutative semigroup.
- Let $(S, *)$ and $(T, \cdot)$ be monoids. Show that $S \times T$ is also a monoid. Show that the identity of $S \times T$ is (e_S, e_T) .
- Let $(S, *)$ and $(T, \cdot)$ be semigroups. Show that the function $f: S \times T \rightarrow S$ defined by $f(s, t) = s$ is a homomorphism of the semigroup $S \times T$ onto the semigroup S .
- Let $(S, *)$ and $(T, \cdot)$ be semigroups. Show that $S \times T$ and $T \times S$ are isomorphic semigroups.
- Prove Theorem 1.

In Exercises 6 through 15, determine whether the relation R on the semigroup S is a congruence relation.

- $S = \mathbb{Z}$ under the operation of ordinary addition; $a R b$ if and only if 2 does not divide $a - b$.
- $S = \mathbb{Z}$ under the operation of ordinary addition; $a R b$ if and only if $a + b$ is even.
- $S =$ any semigroup; $a R b$ if and only if $a = b$.
- $S =$ the set of all rational numbers under the operation of addition; $a/b R c/d$ if and only if $ad = bc$.
- $S =$ the set of all rational numbers under the operation of multiplication; $a/b R c/d$ if and only if $ad = bc$.
- $S = \mathbb{Z}$ under the operation of ordinary addition; $a R b$ if and only if $a \equiv b \pmod{3}$.

- $S = \mathbb{Z}$ under the operation of ordinary addition; $a R b$ if and only if a and b are both even or a and b are both odd.
- $S = \mathbb{Z}^+$ under the operation of ordinary multiplication; $a R b$ if and only if $|a - b| \leq 2$.
- $A = \{0, 1\}$ and $S = A^*$, the free semigroup generated by A under the operation of concatenation; $\alpha R \beta$ if and only if α and β both have an even number of 1's or both have an odd number of 1's.
- $S = \{0, 1\}$ under the operation $*$ defined by the table

$*$	0	1
0	0	1
1	1	0

$a R b$ if and only if $a * a = b * b$. (Hint: Observe that if x is any element in S , then $x * x = 0$.)

- Show that the intersection of two congruence relations on a semigroup is a congruence relation.
- Show that the composition of two congruence relations on a semigroup need not be a congruence relation.
- Describe the quotient semigroup for S and R given in Exercise 10.
- Describe the quotient semigroup for S and R given in Exercise 11.
- Describe the quotient semigroup for S and R given in Exercise 12.
- Describe the quotient semigroup for $S = \mathbb{Z}$ with ordinary addition and R defined by $a R b$ if and only if $a \equiv b \pmod{5}$.

22. Consider the semigroup $S = \{a, b, c, d\}$ with the following operation table.

*	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	a	b
d	d	c	b	a

Consider the congruence relation $R = \{(a, a), (a, b), (b, a), (b, b), (c, c), (c, d), (d, c), (d, d)\}$ on S .

- (a) Write the operation table of the quotient semigroup S/R .
 (b) Describe the natural homomorphism $f_R: S \rightarrow S/R$.
 23. Consider the monoid $S = \{e, a, b, c\}$ with the following operation table.

*	e	a	b	c
e	e	a	b	c
a	a	e	b	c
b	b	c	b	c
c	c	b	b	c

Consider the congruence relation $R = \{(e, e), (e, a), (a, e), (a, a), (b, b), (b, c), (c, b), (c, c)\}$ on S .

- (a) Write the operation table of the quotient monoid S/R .
 (b) Describe the natural homomorphism $f_R: S \rightarrow S/R$.
 24. Let $A = \{0, 1\}$ and consider the free semigroup A^* generated by A under the operation of catenation. Let N be the semigroup of all nonnegative integers under the operation of ordinary addition.
 (a) Verify that the function $f: A^* \rightarrow N$, defined by $f(\alpha) =$ the number of digits in α , is a homomorphism.
 (b) Let R be the following relation on A^* : $\alpha R \beta$ if and only if $f(\alpha) = f(\beta)$. Show that R is a congruence relation on A^* .
 (c) Show that A^*/R and N are isomorphic.
 25. Describe the strategy of the proof of Theorem 4. Outline the proof.

9.4 GROUPS

In this section we examine a special type of monoid, called a group, that has applications in every area where symmetry occurs. Applications of groups can be found in mathematics, physics, and chemistry, as well as in less obvious areas such as sociology. Recent and exciting applications of group theory have arisen in fields such as particle physics and in the solutions of puzzles such as Rubik's cube. In this book, we shall present an important application of group theory to binary codes in Section 11.2.

A group $(G, *)$ is a monoid, with identity e , that has the additional property that for every element $a \in G$ there exists an element $a' \in G$ such that $a * a' = a' * a = e$. Thus a group is a set together with a binary operation $*$ on G such that

- $(a * b) * c = a * (b * c)$ for any elements a, b , and c in G .
- There is a unique element e in G such that

$$a * e = e * a \quad \text{for any } a \in G.$$

- For every $a \in G$, there is an element $a' \in G$, called an *inverse* of a , such that

$$a * a' = a' * a = e.$$

Observe that if $(G, *)$ is a group, then $*$ is a binary operation, so G must be closed under $*$; that is,

$$a * b \in G \quad \text{for any elements } a \text{ and } b \text{ in } G.$$

To simplify our notation, from now on when only one group $(G, *)$ is under consideration and there is no possibility of confusion, we shall write the product $a * b$ of the elements a and b in the group $(G, *)$ simply as ab , and we shall refer to $(G, *)$ simply as G .

A group G is said to be **Abelian** if $ab = ba$ for all elements a and b in G .

EXAMPLE 1

The set of all integers Z with the operation of ordinary addition is an Abelian group. If $a \in Z$, then an inverse of a is its opposite $-a$. ■

EXAMPLE 2

The set Z^+ under the operation of ordinary multiplication is not a group since, for example, the element 2 in Z^+ has no inverse. However, this set together with the given operation is a monoid. ■

EXAMPLE 3

The set of all nonzero real numbers under the operation of ordinary multiplication is a group. An inverse of $a \neq 0$ is $1/a$. ■

EXAMPLE 4

Let G be the set of all nonzero real numbers and let

$$a * b = \frac{ab}{2}.$$

Show that $(G, *)$ is an Abelian group.

Solution We first verify that $*$ is a binary operation. If a and b are elements of G , then $ab/2$ is a nonzero real number and hence is in G . We next verify associativity. Since

$$(a * b) * c = \left(\frac{ab}{2}\right) * c = \frac{(ab)c}{4}$$

and since

$$a * (b * c) = a * \left(\frac{bc}{2}\right) = \frac{a(bc)}{4} = \frac{(ab)c}{4},$$

the operation $*$ is associative.

The number 2 is the identity in G , for if $a \in G$, then

$$a * 2 = \frac{(a)(2)}{2} = a = \frac{(2)(a)}{2} = 2 * a.$$

Finally, if $a \in G$, then $a' = 4/a$ is an inverse of a , since

$$a * a' = a * \frac{4}{a} = \frac{a(4/a)}{2} = 2 = \frac{(4/a)(a)}{2} = \frac{4}{a} * a = a' * a.$$

Since $a * b = b * a$ for all a and b in G , we conclude that G is an Abelian group. ■

Before proceeding with additional examples of groups, we develop several important properties that are satisfied in any group G .

Theorem 1

Let G be a group. Each element a in G has only one inverse in G . ■

Proof Let a' and a'' be inverses of a . Then

$$a'(aa'') = a'e = a'$$

and

$$(a'a)a'' = ea'' = a''.$$

Hence, by associativity,

$$a' = a''.$$

From now on we shall denote the inverse of a by a^{-1} . Thus in a group G we have

$$aa^{-1} = a^{-1}a = e.$$

Theorem 2 Let G be a group and let a, b , and c be elements of G . Then

- (a) $ab = ac$ implies that $b = c$ (**left cancellation property**).
- (b) $ba = ca$ implies that $b = c$ (**right cancellation property**).

Proof

- (a) Suppose that

$$ab = ac.$$

Multiplying both sides of this equation by a^{-1} on the left, we obtain

$$a^{-1}(ab) = a^{-1}(ac)$$

$$(a^{-1}a)b = (a^{-1}a)c \quad \text{by associativity}$$

$$eb = ec \quad \text{by the definition of an inverse}$$

$$b = c \quad \text{by definition of an identity.}$$

- (b) The proof is similar to that of part (a).

Theorem 3 Let G be a group and let a and b be elements of G . Then

- (a) $(a^{-1})^{-1} = a$.
- (b) $(ab)^{-1} = b^{-1}a^{-1}$.

Proof

- (a) We show that a acts as an inverse for a^{-1} :

$$a^{-1}a = aa^{-1} = e.$$

Since the inverse of an element is unique, we conclude that $(a^{-1})^{-1} = a$.

- (b) We easily verify that

$$(ab)(b^{-1}a^{-1}) = a(b(b^{-1}a^{-1})) = a((bb^{-1})a^{-1}) = a(ea^{-1}) = aa^{-1} = e$$

and, similarly,

$$(b^{-1}a^{-1})(ab) = e,$$

so

$$(ab)^{-1} = b^{-1}a^{-1}.$$

Theorem 4 Let G be a group, and let a and b be elements of G . Then

- (a) The equation $ax = b$ has a unique solution in G .
 (b) The equation $ya = b$ has a unique solution in G .

Proof

- (a) The element $x = a^{-1}b$ is a solution of the equation $ax = b$, since

$$a(a^{-1}b) = (aa^{-1})b = eb = b.$$

Suppose now that x_1 and x_2 are two solutions of the equation $ax = b$. Then

$$ax_1 = b \quad \text{and} \quad ax_2 = b.$$

Hence

$$ax_1 = ax_2.$$

Theorem 2 implies that $x_1 = x_2$.

- (b) The proof is similar to that of part (a). ▼

From our discussion of monoids, we know that if a group G has a finite number of elements, then its binary operation can be given by a table, which is generally called a **multiplication table**. The multiplication table of a group $G = \{a_1, a_2, \dots, a_n\}$ under the binary operation $*$ must satisfy the following properties:

1. The row labeled by e must be

$$a_1, a_2, \dots, a_n$$

and the column labeled by e must be

$$a_1$$

$$a_2$$

$$\vdots$$

$$a_n.$$

2. From Theorem 4, it follows that each element b of the group must appear exactly once in each row and column of the table. Thus each row and column is a permutation of the elements $a_1, a_2, \dots, a_n$ of G , and each row (and each column) determines a different permutation.

If G is a group that has a finite number of elements, we say that G is a **finite group**, and the **order** of G is the number of elements $|G|$ in G . We shall now determine the multiplication tables of all nonisomorphic groups of orders 1, 2, 3, and 4.

If G is a group of order 1, then $G = \{e\}$, and we have $ee = e$. Now let $G = \{e, a\}$ be a group of order 2. Then we obtain a multiplication table (Table 9.1) where we need to fill in the blank. The blank can be filled in by e or by a . Since there can be no repeats in any row or column, we must write e in the blank. The multiplication table shown in Table 9.2 satisfies the associativity property and the other properties of a group, so it is the multiplication table of a group of order 2.

Table 9.1

	e	a
e	e	a
a		

Table 9.2

	e	a
e	e	a
a	a	e

Table 9.3

	e	a	b
e	e	a	b
a	a	a	b
b	b		

Table 9.4

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

Next, let $G = \{e, a, b\}$ be a group of order 3. We have a multiplication table (Table 9.3) where we must fill in four blanks. A little experimentation shows that we can only complete the table as shown in Table 9.4. It can be shown (a tedious task) that Table 9.4 satisfies the associative property and the other properties of a group. Thus it is the multiplication table of a group of order 3. Observe that the groups of orders 1, 2, and 3 are also Abelian and that there is just one group of each order for a fixed labeling of the elements.

We next come to a group $G = \{e, a, b, c\}$ of order 4. It is not difficult to show that the possible multiplication table for G can be completed as shown in Tables 9.5 through 9.8. It can be shown that each of these tables satisfies the associative property and the other properties of a group. Thus there are four possible multiplication tables for a group of order 4. Again, observe that a group of order 4 is Abelian. We shall return to groups of order 4 toward the end of this section, where we shall see that there are only two and not four different nonisomorphic groups of order 4.

Table 9.5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table 9.6

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

Table 9.7

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Table 9.8

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

EXAMPLE 5

Let $B = \{0, 1\}$, and let $+$ be the operation defined on B as follows:

$+$	0	1
0	0	1
1	1	0

Then B is a group. In this group, every element is its own inverse.

We next turn to an important example of a group.

EXAMPLE 6

Consider the equilateral triangle shown in Figure 9.3 with vertices 1, 2, and 3. A **symmetry** of the triangle (or of any geometrical figure) is a one-to-one correspondence from the set of points forming the triangle (the geometrical figure) to itself that preserves the distance between adjacent points. Since the triangle is determined by its vertices, a symmetry of the triangle is merely a permutation of the vertices that preserves the distance between adjacent points. Let l_1 , l_2 , and l_3 be the angles

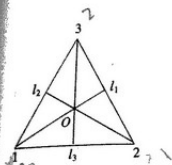


Figure 9.3

bisectors of the corresponding angles as shown in Figure 9.3, and let O be their point of intersection.

We now describe the symmetries of this triangle. First, there is a counterclockwise rotation f_2 of the triangle about O through 120° . Then f_2 can be written (see Section 5.3) as the permutation

$$f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}.$$

We next obtain a counterclockwise rotation f_3 about O through 240° , which can be written as the permutation

$$f_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

Finally, there is a counterclockwise rotation f_1 about O through 360° , which can be written as the permutation

$$f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Of course, f_1 can also be viewed as the result of rotating the triangle about O through 0° .

We may also obtain three additional symmetries of the triangle, g_1 , g_2 , and g_3 , by reflecting about the lines l_1 , l_2 , and l_3 , respectively. We may denote these reflections as the following permutations:

$$g_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \quad g_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \quad g_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}.$$

Observe that the set of all symmetries of the triangle is described by the set of permutations of the set $\{1, 2, 3\}$, which is considered in Section 5.3 and is denoted by S_3 . Thus

$$S_3 = \{f_1, f_2, f_3, g_1, g_2, g_3\}.$$

We now introduce the operation $*$, followed by, on the set S_3 , and we obtain the multiplication table shown in Table 9.9.

Table 9.9

$*$	f_1	f_2	f_3	g_1	g_2	g_3
f_1	f_1	f_2	f_3	g_1	g_2	g_3
f_2	f_2	f_3	f_1	g_3	g_1	g_2
f_3	f_3	f_1	f_2	g_2	g_3	g_1
g_1	g_1	g_2	g_3	f_1	f_2	f_3
g_2	g_2	g_3	g_1	f_3	f_1	f_2
g_3	g_3	g_1	g_2	f_2	f_3	f_1

Each of the entries in this table can be obtained in one of two ways: algebraically or geometrically. For example, suppose that we want to compute $f_2 * g_2$. Geometrically, we proceed as in Figure 9.4. Note that "followed by" here refers to the geometric order. To compute $f_2 * g_2$ algebraically, we compute $f_2 \circ g_2$.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = g_1$$

and find that $f_2 * g_2 = g_1$.

Since composition of functions is always associative, we see that $*$ is an associative operation on S_3 . Observe that f_1 is the identity in S_3 and that every element of S_3 has a unique inverse in S_3 . For example, $f_2^{-1} = f_3$. Hence S_3 is a group called the **group of symmetries of the triangle**. Observe that S_3 is the first example that we have given of a group that is not Abelian.

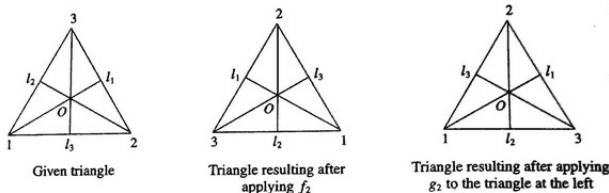


Figure 9.4

EXAMPLE 7

The set of all permutations of n elements is a group of order $n!$ under the operation of composition. This group is called the **symmetric group on n letters** and is denoted by S_n . We have seen that S_3 also represents the group of symmetries of the equilateral triangle.

As in Example 6, we can also consider the group of symmetries of a square. However, it turns out that this group is of order 8, so it does not agree with the group S_4 , whose order is $4! = 24$.

EXAMPLE 8

In Section 9.3 we discussed the monoid Z_n . We now show that Z_n is a group as follows. Let $[a] \in Z_n$. Then we may assume that $0 \leq a < n$. Moreover, $[n - a] \in Z_n$ and since

$$[a] \oplus [n - a] = [a + n - a] = [n] = [0],$$

we conclude that $[n - a]$ is the inverse of $[a]$. Thus, if n is 6, then $[2]$ is the inverse of $[4]$. Observe that Z_n is an Abelian group.

We next turn to a discussion of important subsets of a group. Let H be a subset of a group G such that

- The identity e of G belongs to H .
- If a and b belong to H , then $ab \in H$.
- If $a \in H$, then $a^{-1} \in H$.

Then H is called a **subgroup** of G . Part (b) says that H is a subsemigroup of G . Thus a subgroup of G can be viewed as a subsemigroup having properties (a) and (c).

Observe that if G is a group and H is a subgroup of G , then H is also a group with respect to the operation in G , since the associative property in G also holds in H .

EXAMPLE 9

Let G be a group. Then G and $H = \{e\}$ are subgroups of G , called the **trivial** subgroups of G . ■

EXAMPLE 10

Consider S_3 , the group of symmetries of the equilateral triangle, whose multiplication table is shown in Table 9.9. It is easy to verify that $H = \{f_1, f_2, f_3\}$ is a subgroup of S_3 . ■

EXAMPLE 11

Let A_n be the set of all even permutations (see Section 5.4) in the group S_n . It can be shown from the definition of even permutation that A_n is a subgroup of S_n , called the **alternating group on n letters**. ■

EXAMPLE 12

Let G be a group and let $a \in G$. Since a group is a monoid, we have already defined, in Section 9.2, a^n for $n \in \mathbb{Z}^+$ as $aa \cdots a$ (n factors), and a^0 as e . If n is a negative integer, we now define a^{-n} as $a^{-1}a^{-1} \cdots a^{-1}$ (n factors). Then, if n and m are any integers, we have

$$a^n a^m = a^{n+m}.$$

It is easy to show that

$$H = \{a^i \mid i \in \mathbb{Z}\}$$

is a subgroup of G . ■

Let $(G, *)$ and $(G', *)$ be two groups. Since groups are also semigroups, we can consider isomorphisms and homomorphisms from $(G, *)$ to $(G', *)$.

Since an isomorphism must be a one-to-one and onto function, it follows that two groups whose orders are unequal cannot possibly be isomorphic.

EXAMPLE 13

Let G be the group of real numbers under addition, and let G' be the group of positive real numbers under multiplication. Let $f: G \rightarrow G'$ be defined by $f(x) = e^x$. We now show that f is an isomorphism.

If $f(a) = f(b)$, so that $e^a = e^b$, then $a = b$. Thus f is one to one. If $c \in G'$, then $\ln c \in G$ and

$$f(\ln c) = e^{\ln c} = c,$$

so f is onto. Finally,

$$f(a+b) = e^{a+b} = e^a e^b = f(a)f(b).$$

Hence f is an isomorphism. ■

EXAMPLE 14

Let G be the symmetric group of n letters, and let G' be the group B defined in Example 5. Let $f: G \rightarrow G'$ be defined as follows: for $p \in G$,

$$f(p) = \begin{cases} 0 & \text{if } p \in A_n \quad (\text{the subgroup of all even permutations in } G) \\ 1 & \text{if } p \notin A_n. \end{cases}$$

Then f is a homomorphism.

EXAMPLE 15

Let G be the group of integers under addition, and let G' be the group Z_n as discussed in Example 8. Let $f: G \rightarrow G'$ be defined as follows: If $m \in G$, then $f(m) = [r]$, where r is the remainder when m is divided by n . We now show that f is a homomorphism of G onto G' .

Let $[r] \in Z_n$. Then we may assume that $0 \leq r < n$, so

$$r = 0 \cdot n + r,$$

which means that the remainder when r is divided by n is r . Hence

$$f(r) = [r]$$

and thus f is onto.

Next, let a and b be elements of G expressed as

$$a = q_1 n + r_1, \quad \text{where } 0 \leq r_1 < n, \text{ and } r_1 \text{ and } q_1 \text{ are integers} \quad (1)$$

$$b = q_2 n + r_2, \quad \text{where } 0 \leq r_2 < n, \text{ and } r_2 \text{ and } q_2 \text{ are integers} \quad (2)$$

so that

$$f(a) = [r_1] \quad \text{and} \quad f(b) = [r_2].$$

Then

$$f(a) + f(b) = [r_1] + [r_2] = [r_1 + r_2].$$

To find $[r_1 + r_2]$, we need the remainder when $r_1 + r_2$ is divided by n . Write

$$r_1 + r_2 = q_3 n + r_3, \quad \text{where } 0 \leq r_3 < n, \text{ and } r_3 \text{ and } q_3 \text{ are integers.}$$

Thus

$$f(a) + f(b) = [r_3].$$

Adding, we have

$$a + b = q_1 n + q_2 n + r_1 + r_2 = (q_1 + q_2 + q_3)n + r_3,$$

so

$$f(a + b) = [r_1 + r_2] = [r_3].$$

Hence

$$f(a + b) = f(a) + f(b),$$

which implies that f is a homomorphism.

Note that when n is 2, f assigns each even integer to $[0]$ and each odd integer to $[1]$.

Theorem 5 Let $(G, *)$ and $(G', *')$ be two groups, and let $f: G \rightarrow G'$ be a homomorphism from G to G' .

- (a) If e is the identity in G and e' is the identity in G' , then $f(e) = e'$.
 (b) If $a \in G$, then $f(a^{-1}) = (f(a))^{-1}$.
 (c) If H is a subgroup of G , then

$$f(H) = \{f(h) \mid h \in H\}$$

is a subgroup of G' .

Proof

- (a) Let $x = f(e)$. Then

$$x *' x = f(e) *' f(e) = f(e * e) = f(e) = x,$$

so $x *' x = x$. Multiplying both sides by x^{-1} on the right, we obtain

$$x = x *' x *' x^{-1} = x *' x^{-1} = e'.$$

Thus $f(e) = e'$.

- (b) $a * a^{-1} = e$, so

$$f(a * a^{-1}) = f(e) = e' \quad \text{by part (a)}$$

or

$$f(a) *' f(a^{-1}) = e' \quad \text{since } f \text{ is a homomorphism.}$$

Similarly,

$$f(a^{-1}) *' f(a) = e'.$$

Hence $f(a^{-1}) = (f(a))^{-1}$.

- (c) This follows from Theorem 4 of Section 9.2 and parts (a) and (b). ▀

EXAMPLE 16

The groups S_3 and Z_6 are both of order 6. However, S_3 is not Abelian and Z_6 is Abelian. Hence they are not isomorphic. Remember that an isomorphism preserves all properties defined in terms of the group operations. ▀

EXAMPLE 17

Earlier in this section we found four possible multiplication tables (Tables 9.5 through 9.8) for a group of order 4. We now show that the groups with multiplication Tables 9.6, 9.7, and 9.8 are isomorphic as follows. Let $G = \{e, a, b, c\}$ be the group whose multiplication table is Table 9.6, and let $G' = \{e', a', b', c'\}$ be the group whose multiplication table is Table 9.7, where we put primes on every entry in this last table. Let $f: G \rightarrow G'$ be defined by $f(e) = e'$, $f(a) = b'$, $f(b) = a'$, $f(c) = c'$. We can then verify that under this renaming of elements the two tables become identical, so the corresponding groups are isomorphic. Similarly, let $G'' = \{e'', a'', b'', c''\}$ be the group whose multiplication table is Table 9.8, where we put double primes on every entry in this last table. Let $g: G \rightarrow G''$ be defined by $g(e) = e''$, $g(a) = c''$, $g(b) = b''$, $g(c) = a''$. We can then verify that under this renaming of elements the two tables become identical, so the corresponding groups are isomorphic. That is, the groups given by Tables 9.6, 9.7, and 9.8 are isomorphic.

Now, how can we be sure that Tables 9.5 and 9.6 do not yield isomorphic groups? Observe that if x is any element in the group determined by Table 9.5, then $x^2 = e$. If the groups were isomorphic, then the group determined by Table 9.6 would have the same property. Since it does not, we conclude that these groups are not isomorphic. Thus there are exactly two nonisomorphic groups of order 4.

The group with multiplication Table 9.5 is called the **Klein 4 group** and it is denoted by V . The one with multiplication Table 9.6, 9.7, or 9.8 is denoted by Z_4 , since a relabeling of the elements of Z_4 results in this multiplication table.

9.4 Exercises

In Exercises 1 through 11, determine whether the set together with the binary operation is a group. If it is a group, determine if it is Abelian; specify the identity and the inverse of a generic element.

1. Z , where $*$ is ordinary multiplication.
2. Z , where $*$ is ordinary subtraction.
3. Q , the set of all rational numbers under the operation of addition.
4. Q , the set of all rational numbers under the operation of multiplication.
5. R , under the operation of multiplication.
6. R , where $a * b = a + b + 2$.
7. Z^+ , under the operation of addition.
8. The real numbers that are not equal to -1 , where $a * b = a + b + ab$.
9. The set of odd integers under the operation of multiplication.
10. The set of all $m \times n$ matrices under the operation of matrix addition.
11. If S is a nonempty set, the set $P(S)$, where $A * B = A \oplus B$. (See Section 1.2.)
12. Let $S = \{x \mid x \text{ is a real number and } x \neq 0, x \neq -1\}$. Consider the following functions $f_i: S \rightarrow S, i = 1, 2, \dots, 6$:

$$f_1(x) = x, \quad f_2(x) = 1 - x, \quad f_3(x) = \frac{1}{x}$$

$$f_4(x) = \frac{1}{1-x}, \quad f_5(x) = 1 - \frac{1}{x}, \quad f_6(x) = \frac{x}{x-1}.$$

Show that $G = \{f_1, f_2, f_3, f_4, f_5, f_6\}$ is a group under the operation of composition. Give the multiplication table of G .

13. Consider S_3 , the group of symmetries of the equilateral triangle, and the group in Exercise 12. Prove or disprove that these two groups are isomorphic.

14. Show that the mapping in Example 14 is a homomorphism.

15. Let G be the group defined in Example 4. Solve the following equations:

$$(a) 3 * x = 4 \quad (b) y * 5 = -2$$

16. Let G be a group with identity e . Show that if $a^2 = e$ for all a in G , then G is Abelian.

17. Consider the square shown in Figure 9.5.

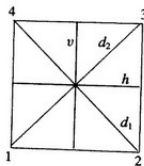


Figure 9.5

The symmetries of the square are as follows:

Rotations f_1, f_2, f_3 , and f_4 through $0^\circ, 90^\circ, 180^\circ$, and 270° , respectively

f_5 and f_6 , reflections about the lines v and h , respectively

f_7 and f_8 , reflections about the diagonals d_1 and d_2 , respectively

Write the multiplication table of D_4 , the group of symmetries of the square.

18. Let G be a group. Show by mathematical induction that if $ab = ba$, then $(ab)^n = a^n b^n$ for $n \in Z^+$.

19. Let G be a finite group with identity e , and let a be an arbitrary element of G . Prove that there exists a nonnegative integer n such that $a^n = e$.
20. Let G be the nonzero integers under the operation of multiplication, and let $H = \{3^n \mid n \in \mathbb{Z}\}$. Is H a subgroup of G ?
21. Let G be the group of integers under the operation of addition, and let $H = \{3k \mid k \in \mathbb{Z}\}$. Is H a subgroup of G ?
22. Let G be an Abelian group with identity e , and let $H = \{x \mid x^2 = e\}$. Show that H is a subgroup of G .
23. Let G be a group, and let $H = \{x \mid x \in G \text{ and } xy = yx \text{ for all } y \in G\}$. Prove that H is a subgroup of G .
24. Let G be a group and let $a \in G$. Define $H_a = \{x \mid x \in G \text{ and } xa = ax\}$. Prove that H_a is a subgroup of G .
25. Let A_n be the set of all even permutations in S_n . Show that A_n is a subgroup of S_n .
26. Let H and K be subgroups of a group G .
 - (a) Prove that $H \cap K$ is a subgroup of G .
 - (b) Show that $H \cup K$ need not be a subgroup of G .
27. Find all subgroups of the group given in Exercise 17.
28. Let G be an Abelian group and n a fixed integer. Prove that the function $f: G \rightarrow G$ defined by $f(a) = a^n$, for $a \in G$, is a homomorphism.
29. Prove that the function $f(x) = |x|$ is a homomorphism from the group G of nonzero real numbers under multiplication to the group G' of positive real numbers under multiplication.
30. Let G be a group with identity e . Show that the function $f: G \rightarrow G$ defined by $f(a) = e$ for all $a \in G$ is a homomorphism.
31. Let G be a group. Show that the function $f: G \rightarrow G$ defined by $f(a) = a^2$ is a homomorphism if and only if G is Abelian.
32. Let G be a group. Show that the function $f: G \rightarrow G$ defined by $f(a) = a^{-1}$ is an isomorphism if and only if G is Abelian.
33. Let G be a group and let a be a fixed element of G . Show that the function $f_a: G \rightarrow G$ defined by $f_a(x) = axa^{-1}$, for $x \in G$, is an isomorphism.
34. Let $G = \{e, a, a^2, a^3, a^4, a^5\}$ be a group under the operation of $a^i a^j = a^r$, where $i + j \equiv r \pmod{6}$. Prove that G and Z_6 are isomorphic.

9.5 PRODUCTS AND QUOTIENTS OF GROUPS

In this section, we shall obtain new groups from other groups by using the ideas of product and quotient. Since a group has more structure than a semigroup, our results will be deeper than analogous results for semigroups as discussed in Section 9.3.

Theorem 1 If G_1 and G_2 are groups, then $G = G_1 \times G_2$ is a group with binary operation defined by

$$(a_1, b_1)(a_2, b_2) = (a_1 a_2, b_1 b_2).$$

Proof By Theorem 1, Section 9.3, we have that G is a semigroup. The existence of an identity and inverses is easy to verify. ▽

EXAMPLE 1

Let G_1 and G_2 be the group Z_2 . For simplicity of notation, we shall write the elements of Z_2 as $\bar{0}$ and $\bar{1}$, respectively, instead of $[0]$ and $[1]$. Then the multiplication table of $G = G_1 \times G_2$ is given in Table 9.10.

Since G is a group of order 4, it must be isomorphic to V or to Z_4 (see Section 9.4), the only groups of order 4. By looking at the multiplication tables, we see that the function $f: V \rightarrow Z_2 \times Z_2$ defined by $f(e) = (\bar{0}, \bar{0})$, $f(a) = (\bar{1}, \bar{0})$, $f(b) = (\bar{0}, \bar{1})$, and $f(c) = (\bar{1}, \bar{1})$ is an isomorphism. ■

Table 9.10 Multiplication Table of $Z_2 \times Z_2$

	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{1})$
$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{1})$
$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$
$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{0})$

If we repeat Example 1 with Z_2 and Z_3 , we find that $Z_2 \times Z_3 \cong Z_6$. It can be shown, in general, that $Z_m \times Z_n \cong Z_{mn}$ if and only if $\text{GCD}(m, n) = 1$, that is, if and only if m and n are relatively prime.

Theorem 1 can obviously be extended to show that if $G_1, G_2, \dots, G_n$ are groups, then $G = G_1 \times G_2 \times \dots \times G_n$ is also a group.

EXAMPLE 2

Let $B = \{0, 1\}$ be the group defined in Example 5 of Section 9.4, where $+$ is defined as follows:

$+$	0	1
0	0	1
1	1	0

Then $B^n = B \times B \times \dots \times B$ (n factors) is a group with operation $\oplus$ defined by

$$(x_1, x_2, \dots, x_n) \oplus (y_1, y_2, \dots, y_n) = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n).$$

The identity of B^n is $(0, 0, \dots, 0)$, and every element is its own inverse. This group is essentially the same as the Boolean algebra B_n defined in Section 6.4, but the binary operation is very different from $\wedge$ and $\vee$.

A congruence relation on a group is simply a congruence relation on the group when it is viewed as a semigroup. We now discuss quotient structures determined by a congruence relation on a group.

Theorem 2

Let R be a congruence relation on the group $(G, *)$. Then the semigroup $(G/R, \otimes)$ is a group, where the operation $\otimes$ is defined on G/R by

$$[a] \otimes [b] = [a * b] \quad (\text{see Section 9.3}).$$

Proof Since a group is a monoid, we know from Corollary 1 of Section 9.3 that G/R is a monoid. We need to show that each element of G/R has an inverse. Let $[a] \in G/R$. Then $[a^{-1}] \in G/R$, and

$$[a] \otimes [a^{-1}] = [a * a^{-1}] = [e].$$

So $[a]^{-1} = [a^{-1}]$. Hence $(G/R, \otimes)$ is a group.

Since the definitions of homomorphism, isomorphism, and congruence for groups involve only the semigroup and monoid structure of groups, the following corollary is an immediate consequence of Theorems 3 and 4 of Section 9.3.

Corollary 1

- (a) If R is a congruence relation on a group G , then the function $f_R: G \rightarrow G/R$, given by $f_R(a) = [a]$, is a group homomorphism.
- (b) If $f: G \rightarrow G'$ is a homomorphism from the group $(G, *)$ onto the group $(G', *)$, and R is the relation defined on G by $a R b$ if and only if $f(a) = f(b)$, for a and b in G , then
- R is a congruence relation.
 - The function $\bar{f}: G/R \rightarrow G'$, given by $\bar{f}([a]) = f(a)$, is an isomorphism from the group $(G/R, \otimes)$ onto the group $(G', *)$. ■

Congruence relations on groups have a very special form, which we will now develop. Let H be a subgroup of a group G , and let $a \in G$. The **left coset** of H in G determined by a is the set $aH = \{ah \mid h \in H\}$. The **right coset** of H in G determined by a is the set $Ha = \{ha \mid h \in H\}$. Finally, we will say that a subgroup H of G is **normal** if $aH = Ha$ for all a in G .

Warning: If $Ha = aH$, it does *not* follow that, for $h \in H$ and $a \in G$, $ha = ah$. It does follow that $ha = ah'$, where h' is some element in H .

If H is a subgroup of G , we shall need in some applications to compute all the left cosets of H in G . First, suppose that $a \in H$. Then $aH \subseteq H$, since H is a subgroup of G ; moreover, if $h \in H$, then $h = ah'$, where $h' = a^{-1}h \in H$, so that $H \subseteq aH$. Thus, if $a \in H$, then $aH = H$. This means that, when finding all the cosets of H , we need not compute aH for $a \in H$, since it will always be H .

EXAMPLE 3

Let G be the symmetric group S_3 discussed in Example 6 of Section 9.4. The subset $H = \{f_1, g_2\}$ is a subgroup of G . Compute all the distinct left cosets of H in G .

Solution If $a \in H$, then $aH = H$. Thus

$$f_1H = g_2H = H.$$

Also,

$$f_2H = \{f_2, g_1\}$$

$$f_3H = \{f_3, g_3\}$$

$$g_1H = \{g_1, f_2\} = f_2H$$

$$g_3H = \{g_3, f_3\} = f_3H.$$

The distinct left cosets of H in G are H , f_2H , and f_3H . ■

EXAMPLE 4

Let G and H be as in Example 3. Then the right coset $Hf_2 = \{f_2, g_3\}$. In Example 3 we saw that $f_2H = \{f_2, g_1\}$. It follows that H is not a normal subgroup of G . ■

EXAMPLE 5

Show that if G is an Abelian group, then every subgroup of G is a normal subgroup.

Solution Let H be a subgroup of G and let $a \in G$ and $h \in H$. Then $ha = ah$, so $Ha = aH$, which implies that H is a normal subgroup of G . ■

Theorem 3 Let R be a congruence relation on a group G , and let $H = [e]$, the equivalence class containing the identity. Then H is a normal subgroup of G and, for each $a \in G$, $[a] = aH = Ha$.

Proof Let a and b be any elements in G . Since R is an equivalence relation, $b \in [a]$ if and only if $[b] = [a]$. Also, G/R is a group by Theorem 2. Therefore $[b] = [a]$ if and only if $[e] = [a]^{-1}[b] = [a^{-1}b]$. Thus $b \in [a]$ if and only if $a^{-1}b \in H$ or $b \in aH$. That is, $b \in [a]$ if and only if $a^{-1}b \in H$ or $b \in aH$. This proves that $[a] = aH$ for every $a \in G$. We can show similarly that $b \in [a]$ if and only if $H = [e] = [b][a]^{-1} = [ba^{-1}]$. This is equivalent to the statement $[a] = Ha$. Thus $[a] = aH = Ha$, and H is normal.

Combining Theorem 3 with Corollary 1, we see that in this case the quotient group G/R consists of all the left cosets of $N = [e]$. The operation in G/R is given by

$$(aN)(bN) = [a] \otimes [b] = [ab] = abN$$

and the function $f_R: G \rightarrow G/R$, defined by $f_R(a) = aN$, is a homomorphism from G onto G/R . For this reason, we will often write G/R as G/N .

We next consider the question of whether every normal subgroup of a group G is the equivalence class of the identity of G for some congruence relation.

Theorem 4 Let N be a normal subgroup of a group G , and let R be the following relation on G :

$$a R b \quad \text{if and only if} \quad a^{-1}b \in N.$$

Then

- R is a congruence relation on G .
- N is the equivalence class $[e]$ relative to R , where e is the identity of G .

Proof

- Let $a \in G$. Then $a R a$, since $a^{-1}a = e \in N$, so R is reflexive. Next suppose that $a R b$, so that $a^{-1}b \in N$. Then $(a^{-1}b)^{-1} = b^{-1}a \in N$, so $b R a$. Hence R is symmetric. Finally, suppose that $a R b$ and $b R c$. Then $a^{-1}b \in N$ and $b^{-1}c \in N$. Then $(a^{-1}b)(b^{-1}c) = a^{-1}c \in N$, so $a R c$. Hence R is transitive. Thus R is an equivalence relation on G .

Next we show that R is a congruence relation on G . Suppose that $a R b$ and $c R d$. Then $a^{-1}b \in N$ and $c^{-1}d \in N$. Since N is normal, $Nd = dN$; that is, for any $n_1 \in N$, $n_1d = dn_2$ for some $n_2 \in N$. In particular, since $a^{-1}b \in N$, we have $a^{-1}bd = dn_2$ for some $n_2 \in N$. Then $(ac)^{-1}bd = (c^{-1}a^{-1})(bd) = c^{-1}(a^{-1}b)d = (c^{-1}d)n_2 \in N$, so $ac R bd$. Hence R is a congruence relation on G .

- Suppose that $x \in N$. Then $x^{-1}e = x^{-1} \in N$ since N is a subgroup, so $x R e$ and therefore $x \in [e]$. Thus $N \subseteq [e]$. Conversely, if $x \in [e]$, then $x R e$, so $x^{-1}e = x^{-1} \in N$. Then $x \in N$ and $[e] \subseteq N$. Hence $N = [e]$.

We see, thanks to Theorems 3 and 4, that if G is any group, then the equivalence classes with respect to a congruence relation on G are always the cosets of

some normal subgroup of G . Conversely, the cosets of any normal subgroup of G are just the equivalence classes with respect to some congruence relation on G . We may now, therefore, translate Corollary 1(b) as follows: Let f be a homomorphism from a group $(G, *)$ onto a group $(G', *)$, and let the kernel of f , $\ker(f)$, be defined by

$$\ker(f) = \{a \in G \mid f(a) = e'\}.$$

Then

- $\ker(f)$ is a normal subgroup of G .
- The quotient group $G/\ker(f)$ is isomorphic to G' .

This follows from Corollary 1 and Theorem 3, since if R is the congruence relation on G given by

$$a R b \text{ if and only if } f(a) = f(b),$$

then it is easy to show that $\ker(f) = [e]$.

EXAMPLE 6

Consider the homomorphism f from Z onto Z_n defined by

$$f(m) = [r],$$

where r is the remainder when m is divided by n . (See Example 15 of Section 9.4.) Find $\ker(f)$.

Solution An integer m in Z belongs to $\ker(f)$ if and only if $f(m) = [0]$, that is, if and only if m is a multiple of n . Hence $\ker(f) = nZ$. ■

9.5 Exercises

- Write the multiplication table for the group $Z_2 \times Z_3$.
- Prove that if G and G' are Abelian groups, then $G \times G'$ is an Abelian group.
- Let G_1 and G_2 be groups. Prove that $G_1 \times G_2$ and $G_2 \times G_1$ are isomorphic.
- Let G_1 and G_2 be groups. Show that the function $f: G_1 \times G_2 \rightarrow G_1$ defined by $f(a, b) = a$, for $a \in G_1$ and $b \in G_2$, is a homomorphism.
- Determine the operational table of the quotient group $Z/3Z$, where Z has operation $+$.
- Let Z be the group of integers under the operation of addition. Prove that the function $f: Z \times Z \rightarrow Z$ defined by $f(a, b) = a + b$ is a homomorphism.
- Let $G = Z_4$. Determine all the left cosets of $H = \{[0]\}$ in G .
- Let $G = Z_4$. Determine all the left cosets of $H = \{[0], [2]\}$ in G .
- Let $G = Z_4$. Determine all the left cosets of $H = \{[0], [1], [2], [3]\}$ in G .
- Let $G = S_3$. Determine all the left cosets of $H = \{f_1, g_1\}$ in G .
- Let $G = S_3$. Determine all the left cosets of $H = \{f_1, g_3\}$ in G .
- Let $G = S_3$. Determine all the left cosets of $H = \{f_1, f_2, f_3\}$ in G .
- Let $G = S_3$. Determine all the left cosets of $H = \{f_1\}$ in G .
- Let $G = S_3$. Determine all the left cosets of $H = \{f_1, f_2, f_3, g_1, g_2, g_3\}$ in G .
- Let $G = Z_8$. Determine all the left cosets of $H = \{[0], [4]\}$ in G .
- Let $G = Z_8$. Determine all the left cosets of $H = \{[0], [2], [4], [6]\}$ in G .

17. Let Z be the group of integers under the operation of addition, and let $G = Z \times Z$. Consider the subgroup $H = \{(x, y) \mid x = y\}$ of G . Describe the left cosets of H in G .
18. Let N be a subgroup of a group G , and let $a \in G$. Define $a^{-1}Na = \{a^{-1}na \mid n \in N\}$.
- Prove that N is a normal subgroup of G if and only if $a^{-1}Na = N$ for all $a \in G$.
19. Let N be a subgroup of group G . Prove that N is a normal subgroup of G if and only if $a^{-1}Na \subseteq N$ for all $a \in G$.
20. Find all the normal subgroups of S_3 .
21. Find all the normal subgroups of D_4 . (See Exercise 17 of Section 9.4.)
22. Let G be a group, and let $H = \{x \mid x \in G \text{ and } xa = ax \text{ for all } a \in G\}$. Show that H is a normal subgroup of G .
23. Let H be a subgroup of a group G . Prove that every left coset aH of H has the same number of elements as H by showing that the function $f_a: H \rightarrow aH$ defined by $f_a(h) = ah$, for $h \in H$, is one to one and onto.
24. Let H and K be normal subgroups of G . Show that $H \cap K$ is a normal subgroup of G .
25. Let G be a group and H a subgroup of G . Let S be the set of all left cosets of H in G , and let T be the set of all right cosets of H in G . Prove that the function $f: S \rightarrow T$ defined by $f(aH) = Ha^{-1}$ is one to one and onto.
26. Let G_1 and G_2 be groups. Let $f: G_1 \times G_2 \rightarrow G_2$ be the homomorphism from $G_1 \times G_2$ onto G_2 given by $f((g_1, g_2)) = g_2$. Compute $\ker(f)$.
27. Let f be a homomorphism from a group G_1 onto a group G_2 , and suppose that G_2 is Abelian. Show that $\ker(f)$ contains all elements of G_1 of the form $aba^{-1}b^{-1}$, where a and b are arbitrary in G_1 .
28. Let G be an Abelian group and N a subgroup of G . Prove that G/N is an Abelian group.
29. Let H be a subgroup of the finite group G and suppose that there are only two left cosets of H in G . Prove that H is a normal subgroup of G .
30. Let H and N be subgroups of the group G . Prove that if N is a normal subgroup of G , then $H \cap N$ is a normal subgroup of H .
31. Let $f: G \rightarrow G'$ be a group homomorphism. Prove that f is one to one if and only if $\ker(f) = \{e\}$.

TIPS FOR PROOFS

The proofs in this chapter are mostly simple direct proofs, in part because we have introduced several new mathematical structures (semigroup, monoids, groups, Abelian groups). With a new structure we first explore the simple consequences of the definitions; for example, Theorem 1, Section 9.2. However, proofs of uniqueness are frequently indirect as in Theorems 1 and 4 in Section 9.4.

The idea of a substructure appears several times in this chapter. In general, to prove that a subset forms a substructure of a mathematical structure, we show that the subset together with the operation(s) satisfy the definition of this type of

structure. But any global property such as associativity is inherited by the subset so we need only check closure properties and properties involving special elements. Thus, to show that a subset is a subgroup, we check closure for the multiplication that the identity belongs to the subset, and that the inverse of each element in the subset belongs to the subset.

Isomorphism is a powerful tool for proving statements; since, roughly speaking, establishing an isomorphism between two structures allows us to transfer knowledge about one structure to the other. This can be seen in Theorem 4, Section 9.2.

KEY IDEAS FOR REVIEW

- Binary operation on A : everywhere defined function $f: A \times A \rightarrow A$
- Commutative binary operation: $a * b = b * a$
- Associative binary operation: $a * (b * c) = (a * b) * c$
- Semigroup: nonempty set S together with an associative binary operation $*$ defined on S
- Monoid: semigroup that has an identity
- Subsemigroup $(T, *)$ of semigroup $(S, *)$: T is a nonempty subset of S and $a * b \in T$ whenever a and b are in T .
- Submonoid $(T, *)$ of monoid $(S, *)$: T is a nonempty subset of S , $e \in T$, and $a * b \in T$ whenever a and b are in T .

- Isomorphism: see page 327
- Homomorphism: see page 329
- Theorem: Let $(S, *)$ and $(T, \circ)$ be monoids with identities e and e' , respectively, and suppose that $f: S \rightarrow T$ is an isomorphism. Then $f(e) = e'$.
- Theorem: If $(S, *)$ and $(T, \circ)$ are semigroups, then $(S \times T, \circ')$ is a semigroup, where $\circ'$ is defined by

$$(s_1, t_1) \circ' (s_2, t_2) = (s_1 * s_2, t_1 \circ t_2).$$

- Congruence relation R on semigroup $(S, *)$: equivalence relation R such that $a R a'$ and $b R b'$ imply that $(a * b) R (a' * b')$
- Theorem: Let R be a congruence relation on the semigroup $(S, *)$. Define the operation $\oplus$ in S/R as follows:

$$[a] \oplus [b] = [a * b].$$

Then $(S/R, \oplus)$ is a semigroup.

- Quotient semigroup or factor semigroup S/R : see page 333
- Z_n : see page 334
- Theorem (fundamental homomorphism theorem): Let $f: S \rightarrow T$ be a homomorphism of the semigroup $(S, *)$ onto the semigroup $(T, \circ)$. Let R be the relation on S defined by $a R b$ if and only if $f(a) = f(b)$, for a and b in S . Then
 - (a) R is a congruence relation.
 - (b) T is isomorphic to S/R .
- Group $(G, *)$: monoid with identity e such that for every $a \in G$ there exists $a' \in G$ with the property that $a * a' = a' * a = e$.
- Theorem: Let G be a group, and let a, b , and c be elements of G . Then

- (a) $ab = ac$ implies that $b = c$ (left cancellation property).
- (b) $ba = ca$ implies that $b = c$ (right cancellation property).

- Theorem: Let G be a group, and let a and b be elements of G . Then
 - (a) $(a^{-1})^{-1} = a$.
 - (b) $(ab)^{-1} = b^{-1}a^{-1}$.
- Order of a group G : $|G|$, the number of elements in G
- S_n : the symmetric group on n letters
- Subgroup: see page 344
- Theorem: Let R be a congruence relation on the group $(G, *)$. Then the semigroup $(G/R, \oplus)$ is a group, where the operation $\oplus$ is defined in G/R by

$$[a] \oplus [b] = [a * b].$$

- Left coset aH of H in G determined by a : $\{ah \mid h \in H\}$
- Normal subgroup: subgroup H such that $aH = Ha$ for all a in G
- Theorem: Let R be a congruence relation on a group G , and let $H = [e]$, the equivalence class containing the identity. Then H is a normal subgroup of G and, for each $a \in G$, $[a] = aH = Ha$.
- Theorem: Let N be a normal subgroup of a group G , and let R be the following relation on G :

$$a R b \quad \text{if and only if} \quad a^{-1}b \in N.$$

Then

- (a) R is a congruence relation on G .
- (b) N is the equivalence class $[e]$ relative to R , where e is the identity of G .

CODING EXERCISES

For each of the following, write the requested program or subroutine in pseudocode (as described in Appendix A) or in a programming language that you know. Test your code either with a paper-and-pencil trace or with a computer run.

Let Z_n be as defined in Section 9.3.

1. Write a function SUM that takes two elements of Z_n , $[x]$ and $[y]$ and returns their sum $[x] \oplus [y]$. The user should be able to input a choice for n .

2. Let $H = \{[0], [2]\}$. Write a subroutine that computes the left cosets of H in Z_6 .
3. Let $H = \{[0], [2], [4], [6]\}$. Write a subroutine that computes the right cosets of H in Z_6 .
4. Write a program that given a finite operation table will determine if the operation satisfies the associative property.
5. Write a program that given a finite group G and a subgroup H determines if H is a normal subgroup of G .

11

GROUPS AND CODING

Prerequisites: Chapter 9

In today's modern world of communication, data items are constantly being transmitted from point to point. This transmission may range from the simple task of a computer terminal interacting with the mainframe computer located 200 feet away, to the more complicated task of sending a signal thousands of miles away via a satellite that is parked in an orbit 20,000 miles from the earth, or to a telephone call or letter to another part of the country. The basic problem in transmission of data is that of receiving the data as sent and not receiving a distorted piece of data. Distortion can be caused by a number of factors.

Coding theory has developed techniques for introducing redundant information in transmitted data that help in detecting, and sometimes in correcting, errors. Some of these techniques make use of group theory. We present a brief introduction to these ideas in this chapter.

11.1 CODING OF BINARY INFORMATION AND ERROR DETECTION

The basic unit of information, called a **message**, is a finite sequence of characters from a finite alphabet. We shall choose as our alphabet the set $B = \{0, 1\}$. Every character or symbol that we want to transmit is now represented as a sequence of m elements from B . That is, every character or symbol is represented in binary form. Our basic unit of information, called a **word**, is a sequence of m 0's and 1's.

The set B is a group under the binary operation $+$ whose table is shown in Table 11.1. (See Example 5 of Section 9.4.)

Table 11.1

$+$	0	1
0	0	1
1	1	0

If we think of B as the group Z_2 , then $+$ is merely mod 2 addition. It follows from Theorem 1 of Section 9.5 that $B^m = B \times B \times \cdots \times B$ (m factors) is a group under

the operation $\oplus$ defined by

$$(x_1, x_2, \dots, x_m) \oplus (y_1, y_2, \dots, y_m) = (x_1 + y_1, x_2 + y_2, \dots, x_m + y_m).$$

This group has been introduced in Example 2 of Section 9.5. Its identity is $\bar{0} = (0, 0, \dots, 0)$ and every element is its own inverse. An element in B^m will be written as $(b_1, b_2, \dots, b_m)$ or more simply as $b_1 b_2 \dots b_m$. Observe that B^m has 2^m elements. That is, the order of the group B^m is 2^m .

Figure 11.1 shows the basic process of sending a word from one point to another point over a transmission channel. An element $x \in B^m$ is sent through the transmission channel and is received as an element $x_t \in B^m$. In actual practice, the transmission channel may suffer disturbances, which are generally called **noise**, due to weather interference, electrical problems, and so on, that may cause a 0 to be received as a 1, or vice versa. This erroneous transmission of digits in a word being sent may give rise to the situation where the word received is different from the word that was sent; that is, $x \neq x_t$. If an error does occur, then x_t could be any element of B^m .

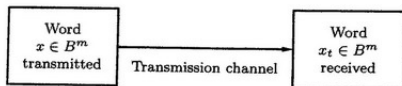


Figure 11.1

The basic task in the transmission of information is to reduce the likelihood of receiving a word that differs from the word that was sent. This is done as follows. We first choose an integer $n > m$ and a one-to-one function $e: B^m \rightarrow B^n$. The function e is called an **(m, n) encoding function**, and we view it as a means of representing every word in B^m as a word in B^n . If $b \in B^m$, then $e(b)$ is called the **code word** representing b . The additional 0's and 1's can provide the means to detect or correct errors produced in the transmission channel.

We now transmit the code words by means of a transmission channel. Then each code word $x = e(b)$ is received as the word x_t in B^n . This situation is illustrated in Figure 11.2.

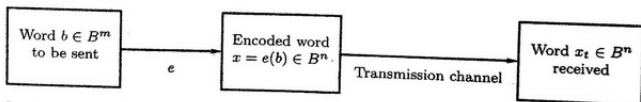


Figure 11.2

Observe that we want an encoding function e to be one to one so that different words in B^m will be assigned different code words.

If the transmission channel is noiseless, then $x_t = x$ for all x in B^n . In this case $x = e(b)$ is received for each $b \in B^m$, and since e is a known function, b may be identified.

In general, errors in transmission do occur. We will say that the code word $x = e(b)$ has been transmitted with **k or fewer errors** if x and x_t differ in at least k but no more than k positions.

Let $e: B^m \rightarrow B^n$ be an (m, n) encoding function. We say that e detects k or fewer errors if whenever $x = e(b)$ is transmitted with k or fewer errors, then x_r is not a code word (thus x_r could not be x and therefore could not have been correctly transmitted). For $x \in B^n$, the number of 1's in x is called the **weight** of x and is denoted by $|x|$.

EXAMPLE 1

Find the weight of each of the following words in B^5 :

- (a) $x = 01000$; (b) $x = 11100$; (c) $x = 00000$; (d) $x = 11111$.

Solution

- (a) $|x| = 1$. (b) $|x| = 3$. (c) $|x| = 0$. (d) $|x| = 5$. ■

EXAMPLE 2

Parity Check Code

The following encoding function $e: B^m \rightarrow B^{m+1}$ is called the **parity $(m, m+1)$ check code**: If $b = b_1b_2 \cdots b_m \in B^m$, define

$$e(b) = b_1b_2 \cdots b_mb_{m+1},$$

where

$$b_{m+1} = \begin{cases} 0 & \text{if } |b| \text{ is even} \\ 1 & \text{if } |b| \text{ is odd.} \end{cases}$$

Observe that b_{m+1} is zero if and only if the number of 1's in b is an even number. It then follows that every code word $e(b)$ has even weight. A single error in the transmission of a code word will change the received word to a word of odd weight and therefore can be detected. In the same way we see that any odd number of errors can be detected.

For a concrete illustration of this encoding function, let $m = 3$. Then

$$\left. \begin{array}{l} e(000) = 0000 \\ e(001) = 0011 \\ e(010) = 0101 \\ e(011) = 0110 \\ e(100) = 1001 \\ e(101) = 1010 \\ e(110) = 1100 \\ e(111) = 1111 \end{array} \right\} \text{code words.}$$

Suppose now that $b = 111$. Then $x = e(b) = 1111$. If the transmission channel transmits x as $x_r = 1101$, then $|x_r| = 3$, and we know that an odd number of errors (at least one) has occurred. ■

It should be noted that if the received word has even weight, then we cannot conclude that the code word was transmitted correctly, since this encoding function does not detect an even number of errors. Despite this limitation, the parity check code is widely used.

EXAMPLE 3

Consider the following $(m, 3m)$ encoding function $e: B^m \rightarrow B^{3m}$. If

$$b = b_1b_2 \cdots b_m \in B^m,$$

define

$$e(b) = e(b_1 b_2 \cdots b_m) = b_1 b_2 \cdots b_m b_1 b_2 \cdots b_m b_1 b_2 \cdots b_m.$$

That is, the encoding function e repeats each word of B^m three times. For a concrete example, let $m = 3$. Then

$$\left. \begin{array}{l} e(000) = 000000000 \\ e(001) = 001001001 \\ e(010) = 010010010 \\ e(011) = 011011011 \\ e(100) = 100100100 \\ e(101) = 101101101 \\ e(110) = 110110110 \\ e(111) = 111111111 \end{array} \right\} \text{code words.}$$

Suppose now that $b = 011$. Then $e(011) = 011011011$. Assume now that the transmission channel makes an error in the underlined digit and that we receive the word 011111011. This is not a code word, so we have detected the error. It is not hard to see that any single error and any two errors can be detected. ■

Let x and y be words in B^m . The **Hamming distance** $\delta(x, y)$ between x and y is the weight, $|x \oplus y|$, of $x \oplus y$. Thus the distance between $x = x_1 x_2 \cdots x_m$ and $y = y_1 y_2 \cdots y_m$ is the number of values of i such that $x_i \neq y_i$, that is, the number of positions in which x and y differ. Using the weight of $x \oplus y$ is a convenient way to count the number of different positions.

EXAMPLE 4

Find the distance between x and y :

- (a) $x = 110110$, $y = 000101$.
 (b) $x = 001100$, $y = 010110$.

Solution

- (a) $x \oplus y = 110011$, so $|x \oplus y| = 4$.
 (b) $x \oplus y = 011010$, so $|x \oplus y| = 3$. ■

Theorem 1
Properties of the
Distance Function

Let x , y , and z be elements of B^m . Then

- (a) $\delta(x, y) = \delta(y, x)$.
 (b) $\delta(x, y) \geq 0$.
 (c) $\delta(x, y) = 0$ if and only if $x = y$.
 (d) $\delta(x, y) \leq \delta(x, z) + \delta(z, y)$.

Proof Properties (a), (b), and (c) are simple to prove and are left as exercises.
 (d) For a and b in B^m ,

$$|a \oplus b| \leq |a| + |b|,$$

since at any position where a and b differ one of them must contain a 1. Also, if $a \in B^m$, then $a \oplus a = \bar{0}$, the identity element in B^m . Then

$$\begin{aligned}\delta(x, y) &= |x \oplus y| = |x \oplus \bar{0} \oplus y| = |x \oplus z \oplus z \oplus y| \\ &\leq |x \oplus z| + |z \oplus y| \\ &= \delta(x, z) + \delta(z, y).\end{aligned}$$

The **minimum distance** of an encoding function $e: B^m \rightarrow B^n$ is the minimum of the distances between all distinct pairs of code words; that is,

$$\min\{\delta(e(x), e(y)) \mid x, y \in B^m\}.$$

EXAMPLE 5

Consider the following (2, 5) encoding function e :

$$\left. \begin{aligned}e(00) &= 00000 \\ e(10) &= 00111 \\ e(01) &= 01110 \\ e(11) &= 11111\end{aligned} \right\} \text{code words.}$$

The minimum distance is 2, as can be checked by computing the minimum of the distances between all six distinct pairs of code words.

 Theorem 2

An (m, n) encoding function $e: B^m \rightarrow B^n$ can detect k or fewer errors if and only if its minimum distance is at least $k + 1$.

Proof Suppose that the minimum distance between any two code words is at least $k + 1$. Let $b \in B^m$, and let $x = e(b) \in B^n$ be the code word representing b . Then x is transmitted and is received as x_i . If x_i were a code word different from x , then $\delta(x, x_i) \geq k + 1$, so x would be transmitted with $k + 1$ or more errors. Thus, if x is transmitted with k or fewer errors, then x_i cannot be a code word. This means that e can detect k or fewer errors.

Conversely, suppose that the minimum distance between code words is $r \leq k$, and let x and y be code words with $\delta(x, y) = r$. If $x_i = y$, that is, if x is transmitted and is mistakenly received as y , then $r \leq k$ errors have been committed and have not been detected. Thus it is not true that e can detect k or fewer errors.

EXAMPLE 6

Consider the (3, 8) encoding function $e: B^3 \rightarrow B^8$ defined by

$$\left. \begin{aligned}e(000) &= 00000000 \\ e(001) &= 10111000 \\ e(010) &= 00101101 \\ e(011) &= 10010101 \\ e(100) &= 10100100 \\ e(101) &= 10001001 \\ e(110) &= 00011100 \\ e(111) &= 00110001\end{aligned} \right\} \text{code words.}$$

How many errors will e detect?

Solution The minimum distance of e is 3, as can be checked by computing the minimum of the distances between all 28 distinct pairs of code words. By Theorem 2, the code will detect k or fewer errors if and only if its minimum distance is at least $k + 1$. Since the minimum distance is 3, we have $3 \geq k + 1$ or $k \leq 2$. Thus the code will detect two or fewer errors. ■

Group Codes

So far, we have not made use of the fact that $(B^n, \oplus)$ is a group. We shall now consider an encoding function that makes use of this property of B^n .

An (m, n) encoding function $e: B^m \rightarrow B^n$ is called a **group code** if

$$e(B^m) = \{e(b) \mid b \in B^m\} = \text{Ran}(e)$$

is a subgroup of B^n .

Recall from the definition of subgroup given in Section 9.4 that N is a subgroup of B^n if (a) the identity of B^n is in N , (b) if x and y belong to N , then $x \oplus y \in N$, and (c) if x is in N , then its inverse is in N . Property (c) need not be checked here, since every element in B^n is its own inverse. Moreover, since B^n is Abelian, every subgroup of B^n is a normal subgroup.

EXAMPLE 7

Consider the $(3, 6)$ encoding function $e: B^3 \rightarrow B^6$ defined by

$$\left. \begin{array}{l} e(000) = 000000 \\ e(001) = 001100 \\ e(010) = 010011 \\ e(011) = 011111 \\ e(100) = 100101 \\ e(101) = 101001 \\ e(110) = 110110 \\ e(111) = 111010 \end{array} \right\} \text{code words.}$$

Show that this encoding function is a group code.

Solution We must show that the set of all code words

$$N = \{000000, 001100, 010011, 011111, 100101, 101001, 110110, 111010\}$$

is a subgroup of B^6 . This is done by first noting that the identity of B^6 belongs to N . Next we verify, by trying all possibilities, that if x and y are elements in N , then $x \oplus y$ is in N . Hence N is a subgroup of B^6 , and the given encoding function is a group code. ■

The strategy of the next proof is similar to the way we often show two sets A and B are the same by showing that $A \subseteq B$ and $B \subseteq A$. Here we show that $\delta = \eta$ by proving $\delta \leq \eta$ and $\eta \leq \delta$.

Theorem 3

Let $e: B^m \rightarrow B^n$ be a group code. The minimum distance of e is the minimum weight of a nonzero code word.

Proof Let δ be the minimum distance of the group code, and suppose that $\delta = \delta(x, y)$, where x and y are distinct code words. Also, let η be the minimum weight of a nonzero code word and suppose that $\eta = |z|$ for a code word z . Since e is a group code, $x \oplus y$ is a nonzero code word. Thus

$$\delta = \delta(x, y) = |x \oplus y| \geq \eta.$$

On the other hand, since 0 and z are distinct code words,

$$\eta = |z| = |z \oplus 0| = \delta(z, 0) \geq \delta.$$

Hence $\eta = \delta$. ▼

One advantage of a group code is given in the following example.

EXAMPLE 8

The minimum distance of the group code in Example 7 is 2, since by Theorem 3 this distance is equal to the smallest number of 1's in any of the seven nonzero code words. To check this directly would require 28 different calculations. ■

We shall now take a brief look at a procedure for generating group codes. First, we need several additional results on Boolean matrices. Consider the set B with operation $+$ defined in Table 11.1. Now let $D = [d_{ij}]$ and $E = [e_{ij}]$ be $m \times n$ Boolean matrices. We define the **mod-2 sum** $D \oplus E$ as the $m \times n$ Boolean matrix $F = [f_{ij}]$, where

$$f_{ij} = d_{ij} + e_{ij}, \quad 1 \leq i \leq m, \quad 1 \leq j \leq n. \quad (\text{Here } + \text{ is addition in } B.)$$

EXAMPLE 9

We have

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \oplus \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} &= \begin{bmatrix} 1+1 & 0+1 & 1+0 & 1+1 \\ 0+1 & 1+1 & 1+0 & 0+1 \\ 1+0 & 0+1 & 0+1 & 1+1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}. \end{aligned}$$

Observe that if $F = D \oplus E$, then f_{ij} is zero when both d_{ij} and e_{ij} are zero or both are one. ■

Table 11.2

$\cdot$	0	1
0	0	0
1	0	1

Next, consider the set $B = \{0, 1\}$ with the binary operation given in Table 11.2. This operation has been seen earlier in a different setting and with a different symbol. In Chapter 6 it was shown that B is the unique Boolean algebra with two elements. In particular, B is a lattice with partial order $\leq$ defined by $0 \leq 0, 0 \leq 1, 1 \leq 1$. Then the reader may easily check that if a and b are any two elements of B ,

$$a \cdot b = a \wedge b \quad (\text{the greatest lower bound of } a \text{ and } b).$$

Thus Table 11.2 for $\cdot$ is just a copy of the table for $\wedge$.

Let $D = [d_{ij}]$ be an $m \times p$ Boolean matrix, and let $E = [e_{ij}]$ be a $p \times n$ Boolean matrix. We define the **mod-2 Boolean product** $D * E$ as the $m \times n$ matrix $F = [f_{ij}]$, where

$$f_{ij} = d_{i1} \cdot e_{1j} + d_{i2} \cdot e_{2j} + \cdots + d_{ip} \cdot e_{pj}, \quad 1 \leq i \leq m, \quad 1 \leq j \leq n.$$

This type of multiplication is illustrated in Figure 11.3. Compare this with similar figures in Section 1.5.

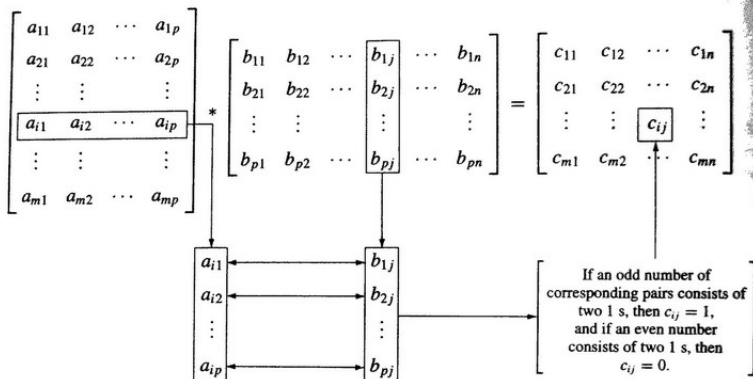


Figure 11.3

EXAMPLE 10

We have

$$\begin{aligned} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} &= \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \end{aligned}$$

The proof of the following theorem is left as an exercise.

Theorem 4 Let D and E be $m \times p$ Boolean matrices, and let F be a $p \times n$ Boolean matrix. Then

$$(D \oplus E) * F = (D * F) \oplus (E * F).$$

That is, a distributive property holds for $\oplus$ and $*$.

We shall now consider the element $x = x_1 x_2 \cdots x_n \in B^n$ as the $1 \times n$ matrix

Theorem 5 Let m and n be nonnegative integers with $m < n$, $r = n - m$, and let $\mathbf{H}$ be an $n \times r$ Boolean matrix. Then the function $f_H: B^n \rightarrow B^r$ defined by

$$f_H(x) = x * \mathbf{H}, \quad x \in B^n$$

is a homomorphism from the group B^n to the group B^r .

Proof If x and y are elements in B^n , then

$$\begin{aligned} f_H(x \oplus y) &= (x \oplus y) * \mathbf{H} \\ &= (x * \mathbf{H}) \oplus (y * \mathbf{H}) \quad \text{by Theorem 4} \\ &= f_H(x) \oplus f_H(y). \end{aligned}$$

Hence f_H is a homomorphism from B^n to B^r .

Corollary 1 Let $m, n, r, \mathbf{H}$, and f_H be as in Theorem 5. Then

$$N = \{x \in B^n \mid x * \mathbf{H} = \bar{0}\}$$

is a normal subgroup of B^n .

Proof It follows from the results in Section 9.5 that N is the kernel of the homomorphism f_H , so it is a normal subgroup of B^n .

Let $m < n$ and $r = n - m$. An $n \times r$ Boolean matrix

$$\mathbf{H} = \begin{Bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{Bmatrix},$$

$n - m = r$ rows

whose last r rows form the $r \times r$ identity matrix, is called a **parity check matrix**. We use $\mathbf{H}$ to define an encoding function $e_H: B^m \rightarrow B^n$. If $b = b_1 b_2 \cdots b_m$, let $x = e_H(b) = b_1 b_2 \cdots b_m x_1 x_2 \cdots x_r$, where

$$\begin{aligned} x_1 &= b_1 \cdot h_{11} + b_2 \cdot h_{21} + \cdots + b_m \cdot h_{m1} \\ x_2 &= b_1 \cdot h_{12} + b_2 \cdot h_{22} + \cdots + b_m \cdot h_{m2} \\ &\vdots \\ x_r &= b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \cdots + b_m \cdot h_{mr}. \end{aligned} \tag{1}$$

Theorem 6 Let $x = y_1 y_2 \cdots y_m x_1 \cdots x_r \in B^n$. Then $x * \mathbf{H} = \bar{0}$ if and only if $x = e_H(b)$ for some $b \in B^m$.

Proof Suppose that $x * \mathbf{H} = \bar{0}$. Then

$$\begin{aligned} y_1 \cdot h_{11} + y_2 \cdot h_{21} + \cdots + y_m \cdot h_{m1} + x_1 &= 0 \\ y_1 \cdot h_{12} + y_2 \cdot h_{22} + \cdots + y_m \cdot h_{m2} + x_2 &= 0 \\ &\vdots \\ y_1 \cdot h_{1r} + y_2 \cdot h_{2r} + \cdots + y_m \cdot h_{mr} + x_r &= 0. \end{aligned}$$

The first equation is of the form

$$a + x_1 = 0, \quad \text{where } a = y_1 \cdot h_{11} + y_2 \cdot h_{21} + \cdots + y_m \cdot h_{m1}.$$

Adding a to both sides, we obtain

$$\begin{aligned} a + (a + x_1) &= a + 0 = a \\ (a + a) + x_1 &= a \\ 0 + x_1 &= a \quad \text{since } a + a = 0 \\ x_1 &= a. \end{aligned}$$

This can be done for each row; therefore,

$$x_i = y_1 \cdot h_{1i} + y_2 \cdot h_{2i} + \cdots + y_m \cdot h_{mi}, \quad 1 \leq i \leq r.$$

Letting $b_1 = y_1, b_2 = y_2, \dots, b_m = y_m$, we see that $x_1, x_2, \dots, x_r$ satisfy the equations in (1). Thus $b = b_1 b_2 \cdots b_m \in B^m$ and $x = e_H(b)$.

Conversely, if $x = e_H(b)$, the equations in (1) can be rewritten by adding x_i to both sides of the i th equation, $i = 1, 2, \dots, n$, as

$$\begin{aligned} b_1 \cdot h_{11} + b_2 \cdot h_{21} + \cdots + b_m \cdot h_{m1} + x_1 &= 0 \\ b_1 \cdot h_{12} + b_2 \cdot h_{22} + \cdots + b_m \cdot h_{m2} + x_2 &= 0 \\ &\vdots \\ b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \cdots + b_m \cdot h_{mr} + x_r &= 0, \end{aligned}$$

which shows $x * \mathbf{H} = \bar{0}$.

Corollary 2 $e_H(B^m) = \{e_H(b) \mid b \in B^m\}$ is a subgroup of B^n .

Proof The result follows from the observation that

$$e_H(B^m) = \ker(f_H)$$

and from Corollary 1. Thus e_H is a group code.

EXAMPLE 11

Let $m = 2, n = 5$, and

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Determine the group code $e_H: B^2 \rightarrow B^5$.

Solution We have $B^2 = \{00, 10, 01, 11\}$. Then

$$e(00) = 00x_1x_2x_3,$$

where x_1, x_2 , and x_3 are determined by the equations in (1). Thus

$$x_1 = x_2 = x_3 = 0$$

and

$$e(00) = 00000.$$

Next

$$e(10) = 10x_1x_2x_3.$$

Using the equations in (1) with $b_1 = 1$ and $b_2 = 0$, we obtain

$$x_1 = 1 \cdot 1 + 0 \cdot 0 = 1$$

$$x_2 = 1 \cdot 1 + 0 \cdot 1 = 1$$

$$x_3 = 1 \cdot 0 + 0 \cdot 1 = 0.$$

Thus $x_1 = 1, x_2 = 1$, and $x_3 = 0$, so

$$e(10) = 10110.$$

Similarly (verify),

$$e(01) = 01011$$

$$e(11) = 11101.$$

11.1 Exercises

Find the weights of the given words.

- (a) 1011 (b) 0110 (c) 1110
- (a) 011101 (b) 11111 (c) 010101
- Consider the (3, 4) parity check code. For each of the received words, determine whether an error will be detected.
 - 0100 (b) 1100
- Consider the (3, 4) parity check code. For each of the received words, determine whether an error will be detected.
 - 0010 (b) 1001
- Consider the (6, 7) parity check code. For each of the received words, determine whether an error will be detected.
 - 1101010 (b) 1010011
 - 0011111 (b) 1001101
- Consider the $(m, 3m)$ encoding function of Example 3, where $m = 4$. For each of the received words, determine whether an error will be detected.
 - 011010011111 (b) 110110010110
- Consider the $(m, 3m)$ encoding function of Example 3, where $m = 4$. For each of the received words, determine whether an error will be detected.
 - 010010110010 (b) 001001110010
- Explain how $|x \oplus y|$ counts the number of positions in which x and y differ.
- Find the distance between x and y .
 - $x = 1100010, y = 1010001$
 - $x = 0100110, y = 0110010$
- Find the distance between x and y .
 - $x = 00111001, y = 10101001$
 - $x = 11010010, y = 00100111$

11. (a) Prove Theorem 1(a).

(b) Prove Theorem 1(b).

12. Prove Theorem 1(c).

13. Find the minimum distance of the (2, 4) encoding function
- e
- .

$$\begin{array}{ll} e(00) = 0000 & e(10) = 0110 \\ e(01) = 1011 & e(11) = 1100 \end{array}$$

14. Find the minimum distance of the (3, 8) encoding function
- e
- .

$$\begin{array}{ll} e(000) = 00000000 & e(100) = 01100101 \\ e(001) = 01110010 & e(101) = 10110000 \\ e(010) = 10011100 & e(110) = 11110000 \\ e(011) = 01110001 & e(111) = 00001111 \end{array}$$

15. Consider the (2, 6) encoding function
- e
- .

$$\begin{array}{ll} e(00) = 000000 & e(10) = 101010 \\ e(01) = 011110 & e(11) = 111000 \end{array}$$

(a) Find the minimum distance of e .(b) How many errors will e detect?

16. Consider the (3, 9) encoding function
- e
- .

$$\begin{array}{ll} e(000) = 000000000 & e(100) = 010011010 \\ e(001) = 011100101 & e(101) = 111101011 \\ e(010) = 010101000 & e(110) = 001011000 \\ e(011) = 110010001 & e(111) = 110000111 \end{array}$$

(a) Find the minimum distance of e .(b) How many errors will e detect?

17. Show that the (2, 5) encoding function
- $e: B^2 \rightarrow B^5$
- defined by

$$\begin{array}{ll} e(00) = 00000 & e(10) = 10101 \\ e(01) = 01110 & e(11) = 11011 \end{array}$$

is a group code.

18. Show that the (3, 7) encoding function
- $e: B^3 \rightarrow B^7$
- defined by

$$\begin{array}{ll} e(000) = 0000000 & e(100) = 1000101 \\ e(001) = 0010110 & e(101) = 1010011 \\ e(010) = 0101000 & e(110) = 1101101 \\ e(011) = 0111110 & e(111) = 1111101 \end{array}$$

is a group code.

19. Find the minimum distance of the group code defined in Exercise 17.

20. Find the minimum distance of the group code defined in Exercise 18.

21. Compute

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \oplus \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

22. Compute

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \oplus \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

23. Compute

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

24. Compute

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

- ★ 25. Let

$$H = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Determine the (2, 5) group code function $e_H: B^2 \rightarrow B^5$.

- ★ 26. Let

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Determine the (3, 6) group code $e_H: B^3 \rightarrow B^6$.

27. Prove Theorem 4.

11.2 DECODING AND ERROR CORRECTION

Consider an (m, n) encoding function $e: B^m \rightarrow B^n$. Once the encoded word $x = e(b) \in B^n$, for $b \in B^m$, is received as the word x_i , we are faced with the problem of identifying the word b that was the original message.

An onto function $d: B^n \rightarrow B^m$ is called an (n, m) **decoding function associated with e** if $d(x_i) = b' \in B^m$ is such that when the transmission channel has no noise, then $b' = b$, that is,

$$d \circ e = 1_{B^m},$$

where 1_{B^m} is the identity function on B^m . The decoding function d is required to be onto so that every received word can be decoded to give a word in B^m . It decodes properly received words correctly, but the decoding of improperly received words may or may not be correct.

EXAMPLE 1

Consider the parity check code that is defined in Example 2 of Section 11.1. We now define the decoding function $d: B^{m+1} \rightarrow B^m$. If $y = y_1 y_2 \cdots y_m y_{m+1} \in B^{m+1}$, then

$$d(y) = y_1 y_2 \cdots y_m.$$

Observe that if $b = b_1 b_2 \cdots b_m \in B^m$, then

$$(d \circ e)(b) = d(e(b)) = b,$$

so $d \circ e = 1_{B^m}$.

For a concrete example, let $m = 4$. Then we obtain $d(10010) = 1001$ and $d(11001) = 1100$. ■

Let e be an (m, n) encoding function and let d be an (n, m) decoding function associated with e . We say that the pair (e, d) **corrects k or fewer errors** if whenever $x = e(b)$ is transmitted correctly or with k or fewer errors and x_i is received, then $d(x_i) = b$. Thus x_i is decoded as the correct message b .

EXAMPLE 2

Consider the $(m, 3m)$ encoding function defined in Example 3 of Section 11.1. We now define the decoding function $d: B^{3m} \rightarrow B^m$. Let

$$y = y_1 y_2 \cdots y_m y_{m+1} \cdots y_{2m} y_{2m+1} \cdots y_{3m}.$$

Then

$$d(y) = z_1 z_2 \cdots z_m,$$

where

$$z_i = \begin{cases} 1 & \text{if } \{y_i, y_{i+m}, y_{i+2m}\} \text{ has at least two 1's} \\ 0 & \text{if } \{y_i, y_{i+m}, y_{i+2m}\} \text{ has less than two 1's.} \end{cases}$$

That is, the decoding function d examines the i th digit in each of the three blocks transmitted. The digit that occurs at least twice in these three blocks is chosen as the decoded i th digit. For a concrete example, let $m = 3$. Then

$$e(100) = 100100100$$

$$e(011) = 011011011$$

$$e(001) = 001001001.$$

Suppose now that $b = 011$. Then $e(011) = 011011011$. Assume now that the transmission channel makes an error in the underlined digit and that we receive the word $x_t = 011111011$. Then, since the first digits in two out of the three blocks are 0, the first digit is decoded as 0. Similarly, the second digit is decoded as 1, since all three second digits in the three blocks are 1. Finally, the third digit is also decoded as 1, for the analogous reason. Hence $d(x_t) = 011$; that is, the decoded word is 011, which is the word that was sent. Therefore, the single error has been corrected. A similar analysis shows that, if e is this $(m, 3m)$ code for any value of m and d is as defined, then (e, d) corrects any single error. ■

Given an (m, n) encoding function $e: B^m \rightarrow B^n$, we often need to determine an (n, m) decoding function $d: B^n \rightarrow B^m$ associated with e . We now discuss a method, called the **maximum likelihood technique**, for determining a decoding function d for a given e .

Since B^m has 2^m elements, there are 2^m code words in B^n . We first list the code words in a fixed order:

$$x^{(1)}, x^{(2)}, \dots, x^{(2^m)}.$$

If the received word is x_t , we compute $\delta(x^{(i)}, x_t)$ for $1 \leq i \leq 2^m$ and choose the first code word, say it is $x^{(s)}$, such that

$$\min_{1 \leq i \leq 2^m} \{\delta(x^{(i)}, x_t)\} = \delta(x^{(s)}, x_t).$$

That is, $x^{(s)}$ is a code word that is closest to x_t and the first in the list. If $x^{(s)} = e(b)$, we define the **maximum likelihood decoding function** d associated with e by

$$d(x_t) = b.$$

Observe that d depends on the particular order in which the code words in $e(B^m)$ are listed. If the code words are listed in a different order, we may obtain a different maximum likelihood decoding function d associated with e .

Theorem 1

Suppose that e is an (m, n) encoding function and d is a maximum likelihood decoding function associated with e . Then (e, d) can correct k or fewer errors if and only if the minimum distance of e is at least $2k + 1$. ■

Proof Assume that the minimum distance of e is at least $2k + 1$. Let $b \in B^m$ and $x = e(b) \in B^n$. Suppose that x is transmitted with k or fewer errors, and x_t is received. This means that $\delta(x, x_t) \leq k$. If z is any other code word, then

$$2k + 1 \leq \delta(x, z) \leq \delta(x, x_t) + \delta(x_t, z) \leq k + d(x_t, z).$$

Thus $\delta(x_t, z) \geq 2k + 1 - k = k + 1$. This means that x is the unique code word that is closest to x_t , so $d(x_t) = b$. Hence (e, d) corrects k or fewer errors.

Conversely, assume that the minimum distance between code words is $r \leq 2k$, and let $x = e(b)$ and $x' = e(b')$ be code words with $\delta(x, x') = r$. Suppose that x' precedes x in the list of code words used to define d . Write $x = b_1 b_2 \dots b_n$, $x' = b'_1 b'_2 \dots b'_n$. Then $b_i \neq b'_i$ for exactly r integers i between 1 and n . Assume, for simplicity, that $b_1 \neq b'_1, b_2 \neq b'_2, \dots, b_r \neq b'_r$, but $b_i = b'_i$ when $i > r$. Any other case is handled in the same way.

- (a) Suppose that $r \leq k$. If x is transmitted as $x_t = x'$, then $r \leq k$ errors have been committed, but $d(x_t) = b'$; so (e, d) has not corrected the r errors.
 (b) Suppose that $k + 1 \leq r \leq 2k$, and let

$$y = b'_1 b'_2 \cdots b'_k b_{k+1} \cdots b_n.$$

If x is transmitted as $x_t = y$, then $\delta(x_t, x') = r - k \leq k$ and $\delta(x_t, x) \geq k$. Thus x' is at least as close to x_t as x is, and x' precedes x in the list of code words; so $d(x_t) \neq b$. Then we have committed k errors, which (e, d) has not corrected. ▼

EXAMPLE 3

Let e be the $(3, 8)$ encoding function defined in Example 6 of Section 11.1, and let d be an $(8, 3)$ maximum likelihood decoding function associated with e . How many errors can (e, d) correct?

Solution Since the minimum distance of e is 3, we have $3 \geq 2k + 1$, so $k \leq 1$. Thus (e, d) can correct one error. ■

We now discuss a simple and effective technique for determining a maximum likelihood decoding function associated with a given group code. First, we prove the following result.

Theorem 2

If K is a finite subgroup of a group G , then every left coset of K in G has exactly as many elements as K . ●

Proof Let aK be a left coset of K in G , where $a \in G$. Consider the function $f: K \rightarrow aK$ defined by

$$f(k) = ak, \quad \text{for } k \in K.$$

We show that f is one to one and onto.

To show that f is one to one, we assume that

$$f(k_1) = f(k_2), \quad k_1, k_2 \in K.$$

Then

$$ak_1 = ak_2.$$

By Theorem 2 of Section 9.4, $k_1 = k_2$. Hence f is one to one.

To show that f is onto, let b be an arbitrary element in aK . Then $b = ak$ for some $k \in K$. We now have

$$f(k) = ak = b,$$

so f is onto. Since f is one to one and onto, K and aK have the same number of elements. ▼

Let $e: B^m \rightarrow B^n$ be an (m, n) encoding function that is a group code. Thus the set N of code words in B^n is a subgroup of B^n whose order is 2^m , say $N = \{x^{(1)}, x^{(2)}, \dots, x^{(2^m)}\}$.

Suppose that the code word $x = e(b)$ is transmitted and that the word x_r is received. The left coset of x_r is

$$x_r \oplus N = \{\epsilon_1, \epsilon_2, \dots, \epsilon_{2^m}\},$$

where $\epsilon_i = x_r \oplus x^{(i)}$. The distance from x_r to code word $x^{(i)}$ is just $|\epsilon_i|$, the weight of ϵ_i . Thus, if ϵ_j is a coset member with smallest weight, then $x^{(j)}$ must be a code word that is closest to x_r . In this case, $x^{(j)} = \bar{0} \oplus x^{(j)} = x_r \oplus x_r \oplus x^{(j)} = x_r \oplus \epsilon_j$. An element ϵ_j , having smallest weight, is called a **coset leader**. Note that a coset leader need not be unique.

If $e: B^m \rightarrow B^n$ is a group code, we now state the following procedure for obtaining a maximum likelihood decoding function associated with e .

STEP 1: Determine all the left cosets of $N = e(B^m)$ in B^n .

STEP 2: For each coset, find a coset leader (a word of least weight). Steps 1 and 2 can be carried out in a systematic tabular manner that will be described later.

STEP 3: If the word x_r is received, determine the coset of N to which x_r belongs. Since N is a normal subgroup of B^n , it follows from Theorems 3 and 4 of Section 9.5 that the cosets of N form a partition of B^n , so each element of B^n belongs to one and only one coset of N in B^n . Moreover, there are $2^n/2^m$ or 2^r distinct cosets of N in B^n .

STEP 4: Let ϵ be a coset leader for the coset determined in Step 3. Compute $x = x_r \oplus \epsilon$. If $x = e(b)$, we let $d(x_r) = b$. That is, we decode x_r as b .

To implement the foregoing procedure, we must keep a complete list of all the cosets of N in B^n , usually in tabular form, with each row of the table containing one coset. We identify a coset leader in each row. Then, when a word x_r is received, we locate the row that contains it, find the coset leader for that row, and add it to x_r . This gives us the code word closest to x_r . We can eliminate the need for these additions if we construct a more systematic table.

Before illustrating with an example, we make several observations. Let

$$N = \{x^{(1)}, x^{(2)}, \dots, x^{(2^m)}\},$$

where $x^{(1)}$ is $\bar{0}$, the identity of B^n .

Steps 1 and 2 in the preceding decoding algorithm are carried out as follows. First, list all the elements of N in a row, starting with the identity $\bar{0}$ at the left. Thus we have

$$\bar{0} \quad x^{(2)} \quad x^{(3)} \quad \dots \quad x^{(2^m)}.$$

This row is the coset $[\bar{0}]$, and it has $\bar{0}$ as its coset leader. For this reason we will also refer to $\bar{0}$ as ϵ_1 . Now choose any word y in B^n that has not been listed in the first row. List the elements of the coset $y \oplus N$ as the second row. This coset also has 2^m elements. Thus we have the two rows

$$\begin{array}{ccccccc} \bar{0} & x^{(2)} & x^{(3)} & \dots & x^{(2^m)} \\ y \oplus \bar{0} & y \oplus x^{(2)} & y \oplus x^{(3)} & \dots & y \oplus x^{(2^m)} \end{array}$$

In the coset $y \oplus N$, pick an element of least weight, a coset leader, which we denote by $\epsilon^{(2)}$. In case of ties, choose any element of least weight. Recall from Section 9.5 that, since $\epsilon^{(2)} \in y \oplus N$, we have $y \oplus N = \epsilon^{(2)} \oplus N$. This means that every word

in the second row can be written as $\epsilon^{(2)} \oplus v$, $v \in N$. Now rewrite the second row as follows:

$$\epsilon^{(2)} \quad \epsilon^{(2)} \oplus x^{(2)} \quad \epsilon^{(2)} \oplus x^{(3)} \quad \dots \quad \epsilon^{(2)} \oplus x^{(2^m)}$$

with $\epsilon^{(2)}$ in the leftmost position.

Next, choose another element z in B^n that has not yet been listed in either of the first two rows and form the third row ($z \oplus x^{(j)}$), $1 \leq j \leq 2^m$ (another coset of N in B^n). This row can be rewritten in the form

$$\epsilon^{(3)} \quad \epsilon^{(3)} \oplus x^{(2)} \quad \epsilon^{(3)} \oplus x^{(3)} \quad \dots \quad \epsilon^{(3)} \oplus x^{(2^m)},$$

where $\epsilon^{(3)}$ is a coset leader for the row.

Continue this process until all elements of B^n have been listed. The resulting Table 11.3 is called a **decoding table**. Notice that it contains 2^r rows, one for each coset of N . If we receive the word x_i , we locate it in the table. If x is the element of N that is at the top of the column containing x_i , then x is the code word closest to x_i . Thus, if $x = e(b)$, we let $d(x_i) = b$.

Table 11.3

$\bar{0}$	$x^{(2)}$	$x^{(3)}$	...	$x^{(2^m-1)}$
$\epsilon^{(2)}$	$\epsilon^{(2)} \oplus x^{(2)}$	$\epsilon^{(2)} \oplus x^{(3)}$	...	$\epsilon^{(2)} \oplus x^{(2^m-1)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
$\epsilon^{(2^r)}$	$\epsilon^{(2^r)} \oplus x^{(2)}$	$\epsilon^{(2^r)} \oplus x^{(3)}$	...	$\epsilon^{(2^r)} \oplus x^{(2^m-1)}$

EXAMPLE 4

Consider the (3, 6) group code defined in Example 7 of Section 11.1. Here

$$\begin{aligned} N &= \{000000, 001100, 010011, 011111, 100101, 101001, 110110, 111010\} \\ &= \{x^{(1)}, x^{(2)}, \dots, x^{(8)}\} \end{aligned}$$

as defined in Example 1. We now implement the decoding procedure for e as follows.

STEPS 1 and 2: Determine all the left cosets of N in B^6 , as rows of a table. For each row i , locate the coset leader ϵ_i , and rewrite the row in the order

$$\epsilon_i, \quad \epsilon_i \oplus 001100, \quad \epsilon_i \oplus 010011, \quad \dots, \quad \epsilon_i \oplus 111010.$$

The result is shown in Table 11.4.

Table 11.4

000000	001100	010011	011111	100101	101001	110110	111010
000001	001101	010010	011110	100100	101000	110111	111011
000010	001110	010001	011101	100111	101011	110100	111000
000100	001000	010111	011011	100001	101101	110010	111110
010000	011100	000011	001111	110101	111001	100110	101010
100000	101100	110011	111111	<u>000101</u>	001001	010110	011010
000110	001010	<u>010101</u>	011001	100011	101111	110000	111100
010100	011000	000111	001011	110001	111101	<u>100010</u>	101110

STEPS 3 and 4: If we receive the word 000101, we decode it by first locating it in the decoding table: it appears in the fifth column, where it is underlined. The word at the top of the fifth column is 100101. Since $e(100) = 100101$, we decode 000101 as 100. Similarly, if we receive the word 010101, we first locate it in the third column of the decoding table, where it is underlined twice. The word at the top of the third column is 010011. Since $e(010) = 010011$, we decode 010101 as 010.

We make the following observations for this example. In determining the decoding table in Steps 1 and 2, there was more than one candidate for coset leader of the last two cosets. In row 7 we chose 00110 as coset leader. If we had chosen 001010 instead, row 7 would have appeared in the rearranged form

$$001010 \quad 001010 \oplus 001100 \quad \dots \quad 001010 \oplus 111010$$

or

$$001010 \quad 000110 \quad 011001 \quad 010101 \quad 101111 \quad 100011 \quad 111100 \quad 110000.$$

The new decoding table is shown in Table 11.5.

Table 11.5

000000	001100	010011	011111	100101	101001	110110	111010
000001	001101	010010	011110	100100	101000	110111	111011
000010	001110	010001	011101	100111	101011	110100	111000
000100	001000	010111	011011	100001	101101	110010	111110
010000	011100	000011	001111	110101	111001	100110	101010
100000	101100	110011	111111	000101	001001	010110	011010
001010	000110	011001	<u>010101</u>	101111	100011	111100	110000
010100	011000	000111	001011	110001	111101	100010	101110

Now, if we receive the word 010101, we first locate it in the *fourth* column of Table 11.5. The word at the top of the fourth column is 011111. Since $e(011) = 011111$, we decode 010101 as 011. ■

Suppose that the (m, n) group code is $e_H: B^m \rightarrow B^n$, where H is a given parity check matrix. In this case, the decoding technique above can be simplified. We now turn to a discussion of this situation.

Recall from Section 11.1 that $r = n - m$,

$$H = \begin{bmatrix} h_{11} & h_{12} & \dots & h_{1r} \\ h_{21} & h_{22} & \dots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \dots & h_{mr} \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

and the function $f_H: B^n \rightarrow B^r$ defined by

$$f_H(x) = x * \mathbf{H}$$

is a homomorphism from the group B^n to the group B^r .

Theorem 3 If $m, n, r, \mathbf{H}$, and f_H are as defined, then f_H is onto.

Proof Let $b = b_1 b_2 \cdots b_r$ be any element in B^r . Letting

$$x = \underbrace{00 \cdots 0}_{m \text{ 0's}} b_1 b_2 \cdots b_r$$

we obtain $x * \mathbf{H} = b$. Thus $f_H(x) = b$, so f_H is onto.

It follows from Corollary 1 of Section 9.5 that B^r and B^n/N are isomorphic, where $N = \ker(f_H) = e_H(B^m)$, under the isomorphism $g: B^n/N \rightarrow B^r$ defined by

$$g(xN) = f_H(x) = x * \mathbf{H}.$$

The element $x * \mathbf{H}$ is called the **syndrome** of x . We now have the following result.

Theorem 4 Let x and y be elements in B^n . Then x and y lie in the same left coset of N in B^n if and only if $f_H(x) = f_H(y)$, that is, if and only if they have the same syndrome.

Proof It follows from Theorem 4 of Section 9.5 that x and y lie in the same left coset of N in B^n if and only if $x \oplus y = (-x) \oplus y \in N$. Since $N = \ker(f_H)$, $x \oplus y \in N$ if and only if

$$\begin{aligned} f_H(x \oplus y) &= \bar{0}_{B^r} \\ f_H(x) \oplus f_H(y) &= \bar{0}_{B^r} \\ f_H(x) &= f_H(y). \end{aligned}$$

In this case, the decoding procedure given previously can be modified as follows. Suppose that we compute the syndrome of each coset leader. If the word x_i is received, we also compute $f_H(x_i)$, the syndrome of x_i . By comparing $f_H(x_i)$ and the syndromes of the coset leaders, we find the coset in which x_i lies. Suppose that a coset leader of this coset is e . We now compute $x = x_i \oplus e$. If $x = e(b)$, we then decode x_i as b . Thus we need only the coset leaders and their syndromes in order to decode. We state the new procedure in detail.

STEP 1: Determine all left cosets of $N = e_H(B^m)$ in B^n .

STEP 2: For each coset, find a coset leader, and compute the syndrome of each leader.

STEP 3: If x_i is received, compute the syndrome of x_i and find the coset leader e having the same syndrome. Then $x_i \oplus e = x$ is a code word $e_H(b)$, and $d(x_i) = b$.

For this procedure, we do not need to keep a table of cosets, and we can avoid the work of computing a decoding table. Simply list all cosets once, in any order, and select a coset leader from each coset. Then keep a table of these coset leaders and their syndromes. The foregoing procedure is easily implemented with such a table.

EXAMPLE 5

Consider the parity check matrix

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and the $(3, 6)$ group $e_H: B^3 \rightarrow B^6$. Then

$$\left. \begin{array}{l} e(000) = 000000 \\ e(001) = 001011 \\ e(010) = 010101 \\ e(011) = 011110 \\ e(100) = 100110 \\ e(101) = 101101 \\ e(110) = 110011 \\ e(111) = 111000 \end{array} \right\} \text{code words.}$$

Table 11.6

Syndrome of Coset Leader	Coset Leader
000	000000
001	000001
010	000010
011	001000
100	000100
101	010000
110	100000
111	001100

Thus

$$N = \{000000, 001011, 010101, 011110, 100110, 101101, 110011, 111000\}.$$

We now implement the decoding procedure as follows.

In Table 11.6 we give only the coset leaders together with their syndromes. Suppose now that we receive the word 001110. We compute the syndrome of $x_r = 001110$, obtaining $f_H(x_r) = x_r * H = 101$, which is the sixth entry in the first column of Table 11.6. This means that x_r lies in the coset whose leader is $\epsilon = 010000$. We compute $x = x_r \oplus \epsilon = 001110 \oplus 010000 = 011110$. Since $e(011) = 011110$, we decode 001110 as 011.

11.2 Exercises

- Let d be the $(4, 3)$ decoding function defined by letting m be 3 in Example 1. Determine $d(y)$ for the word y in B^4 .
(a) $y = 0110$ (b) $y = 1011$
- Let d be the $(6, 5)$ decoding function defined by letting m be 5 in Example 1. Determine $d(y)$ for the word y in B^6 .
(a) $y = 001101$ (b) $y = 110100$

- Let d be the $(6, 2)$ decoding function defined in Example 2. Determine $d(y)$ for the word y in B^6 .
(a) $y = 111011$ (b) $y = 010100$
- Let d be the $(9, 3)$ decoding function defined in the same way as the decoding function in Example 2. Determine $d(y)$ for the word y in B^9 .
(a) $y = 101111101$ (b) $y = 100111100$

In Exercises 5 through 10, let e be the indicated encoding function and let d be an associated maximum likelihood decoding function. Determine the number of errors that (e, d) will correct.

5. e is the encoding function in Exercise 13 of Section 11.1.
6. e is the encoding function in Exercise 14 of Section 11.1.
7. e is the encoding function in Exercise 15 of Section 11.1.
8. e is the encoding function in Exercise 16 of Section 11.1.
9. e is the encoding function in Exercise 17 of Section 11.1.
10. e is the encoding function in Exercise 18 of Section 11.1.
11. Consider the group code defined in Exercise 17 of Section 11.1. Decode the following words relative to a maximum likelihood decoding function.
 - (a) 11110 (b) 10011 (c) 10100

12. Consider the $(2, 4)$ group encoding function $e: B^2 \rightarrow B^4$ defined by

$$\begin{array}{ll} e(00) = 0000 & e(10) = 1001 \\ e(01) = 0111 & e(11) = 1111. \end{array}$$

Decode the following words relative to a maximum likelihood decoding function.

- (a) 0011 (b) 1011 (c) 1111

13. Consider the $(3, 5)$ group encoding function $e: B^3 \rightarrow B^5$ defined by

$$\begin{array}{ll} e(000) = 00000 & e(100) = 10011 \\ e(001) = 00110 & e(101) = 10101 \\ e(010) = 01001 & e(110) = 11010 \\ e(011) = 01111 & e(111) = 11100. \end{array}$$

Decode the following words relative to a maximum likelihood decoding function.

- (a) 11001 (b) 01010 (c) 00111

14. Consider the $(3, 6)$ group encoding function $e: B^3 \rightarrow B^6$ defined by

$$\begin{array}{ll} e(000) = 000000 & e(100) = 100101 \\ e(001) = 000110 & e(101) = 100011 \\ e(010) = 010010 & e(110) = 110111 \\ e(011) = 010100 & e(111) = 110001. \end{array}$$

Decode the following words relative to a maximum likelihood decoding function.

- (a) 011110 (b) 101011 (c) 110010

15. Let G be a group and H a subgroup of G .

- (a) Prove that if g_1 and g_2 are elements of G , then either $g_1H = g_2H$ or $g_1H \cap g_2H = \{ \}$.
- (b) Use the result of part (a) to show that the left cosets of H form a partition of G .

In Exercises 16 through 18, determine the coset leaders for $N = e_H(B^m)$ for the given parity check matrix H .

$$16. H = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$17. H = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$18. H = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In Exercises 19 through 21, compute the syndrome for each coset leader found in the specified exercise.

19. Exercise 16.

20. Exercise 17.

21. Exercise 18.

22. Let

$$H = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

be a parity check matrix. Decode the following words relative to a maximum likelihood decoding function.

- (a) 0101 (b) 1010 (c) 1101

23. Let

$$H = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Decode the following words relative to a maximum likelihood decoding function associated with e_H .

- (a) 10100 (b) 01101 (c) 11011

24. Let

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Decode the following words relative to a maximum likelihood decoding function associated with e_H .

- (a) 011001 (b) 101011 (c) 111010

TIPS FOR PROOFS

The proofs in this chapter rely heavily on earlier results. Many of the concepts developed throughout the book are applied here to problems of coding and decoding. In Section 11.1, we emphasize the similarity of proving two numbers equal to proving two sets are the same. Analogous proofs could be developed for any relation that has the antisymmetric property as

"is less than" and "is a subset of" do.

In Section 11.2, Theorem 2 we use a one-to-one, onto function to "match" the elements of two sets in order to show that they have the same number of elements. This is also a technique that can be used in solving counting problems if the cardinality of one of the sets used is known.

KEY IDEAS FOR REVIEW

- Message: finite sequence of characters from a finite alphabet
- Word: sequence of 0's and 1's
- (m, n) encoding function: one-to-one function $e: B^m \rightarrow B^n, m < n$
- Code word: element in $\text{Ran}(e)$
- Weight of $x, |x|$: number of 1's in x
- Parity check code: see page 403
- Hamming distance between x and $y, \delta(x, y): |x \oplus y|$
- Theorem (Properties of the Distance Function): Let x, y , and z be elements of B^n . Then
 - (a) $\delta(x, y) = \delta(y, x)$.
 - (b) $\delta(x, y) \geq 0$.
 - (c) $\delta(x, y) = 0$ if and only if $x = y$.
 - (d) $\delta(x, y) \leq \delta(x, z) + \delta(z, y)$.
- Minimum distance of an (m, n) encoding function: minimum of the distances between all distinct pairs of code words
- Theorem: An (m, n) encoding function $e: B^m \rightarrow B^n$ can detect k or fewer errors if and only if its minimum distance is at least $k + 1$.
- Group code: (m, n) encoding function $e: B^m \rightarrow B^n$ such that $e(B^m) = \{e(b) \mid b \in B^m\}$ is a subgroup of B^n
- Theorem: The minimum distance of a group code is the minimum weight of a nonzero code word.
- Mod-2 sum of Boolean matrices D and $E, D \oplus E$: see page 407
- Mod-2 Boolean product of Boolean matrices D and $E, D * E$: see page 408
- Theorem: Let m and n be nonnegative integers with $m < n$, $r = n - m$, and let H be an $n \times r$ Boolean matrix. Then the function $f_H: B^n \rightarrow B^r$ defined by

$$f_H(x) = x * H, \quad x \in B^n$$
 is a homomorphism from the group B^n to the group B^r .
- Group code e_H corresponding to parity check matrix H : see page 409
- (n, m) decoding function: see page 413
- Maximum likelihood decoding function associated with e : see page 414
- Theorem: Suppose that e is an (m, n) encoding function and d is a maximum likelihood decoding function associated with e . Then (e, d) can correct k or fewer errors if and only if the minimum distance is at least $2k + 1$.
- Decoding procedure for a group code: see page 416
- Decoding procedure for a group code given by a parity check matrix: see page 419

CODING EXERCISES

For each of the following, write the requested program or subroutine in pseudocode (as described in Appendix A) or in a programming language that you know. Test your code either with a paper-and-pencil trace or with a computer run.

1. Write a function that finds the weight of a word in B^n .
2. Write a subroutine that computes the Hamming distance between two words in B^n .
3. Let M and N be Boolean matrices of size $n \times n$. Write a

program that given M and N returns their mod-2 Boolean product.

4. Write a subroutine to simulate the $(m, 3m)$ -encoding function $e: B^m \rightarrow B^{3m}$ described in Example 3, Section 11.1.
5. Write a subroutine to simulate the decoding function d for the encoding function of Exercise 4 as described in Example 2, Section 11.2.

CHAPTER 11 SELF-TEST

1. Consider the $(3, 4)$ parity check code. For each of the received words, determine whether an error will be detected.
(a) 1101 (b) 1010 (c) 1111 (d) 0011

2. Consider the $(m, 3m)$ encoding function with $m = 4$. For each of the received words, determine whether an error will be detected.

- (a) 001100100011 (b) 110111001101
(c) 010111010011

3. Let e be the $(3, 5)$ encoding function defined by

$$\begin{array}{ll} e(000) = 0000 & e(100) = 01010 \\ e(001) = 11110 & e(101) = 10100 \\ e(010) = 01101 & e(110) = 00111 \\ e(011) = 10011 & e(111) = 11001. \end{array}$$

How many errors will e detect?

4. Show that the $(3, 5)$ encoding function in Problem 3 is a group code.
5. Let e be the encoding function defined in Problem 3 and let d be an associated maximum likelihood decoding function. Determine the number of errors that (e, d) will correct.
6. Let

$$H = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

be a parity check matrix. Decode 0110 relative to a maximum likelihood decoding function associated with e_H .